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Anomaly Detection in Cryptocurrency Transactions Using Machine Learning

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Abstract

Cryptocurrencies have gained widespread popularity recently, with Bitcoin being one of the most prominent examples. As the adoption of cryptocurrencies continues to grow, so does the importance of ensuring the security and integrity of blockchain networks. This thesis focuses on the critical task of anomaly detection in Bitcoin transaction data using machine learning techniques.

The initial objective of this research is to develop an effective and robust framework for identifying anomalous Bitcoin transaction blocks. To achieve this purpose, we employ a two-step approach. At first, we utilize unsupervised K-means clustering to categorize blocks of Bitcoin transaction data into two distinct groups: normal and suspicious. This clustering step lets us label blocks based on their transaction patterns.

Subsequently, we apply supervised algorithms such as Principal Component Analysis (PCA) and Random Forest to refine and enhance the anomaly detection model. PCA allows us to reduce the dimensionality of the data, thereby improving the model's efficiency while preserving essential information. Random Forest, a powerful ensemble learning technique, is leveraged to train a robust classifier to identify anomalies in Bitcoin transaction blocks accurately.

This research not only aids in safeguarding the integrity of cryptocurrency transactions but also lays the foundation for further study in forecasting and preventing potential threats and attacks within the cryptocurrency ecosystem. As cryptocurrency transactions play a vital role in the global financial landscape, developing advanced anomaly detection techniques becomes essential for maintaining trust and security in blockchain networks.

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1 CHAPTER 1: INTRODUCTION

The emergence of suspicious activities within network structures is a historical problem as old as these systems' inception. Such activities stemming from entities or behaviours deviating from the expected norms are commonly labelled anomalies. Anomaly detection has found broad application in the realm of blockchain, where its primary mission is to identify and address many security concerns, ranging from financial fraud and network intrusion to anti-money laundering measures. Carlozo (2017) elaborates that a blockchain is a decentralized and distributed ledger that continuously updates and secures its records, preventing the inclusion of fraudulent information and maintaining the immutability of previously recorded data. Despite these safeguards, participants within a blockchain network may attempt illicit activities, occasionally managing to exploit vulnerabilities for their benefit. In some instances, individuals can manipulate the system for personal gain, even within the framework of a blockchain.

Cutting-edge anomaly detection methods have traditionally been developed for centralized systems. However, as blockchain technology advances, there is a growing demand for implementing anomaly detection procedures within these decentralized systems. This thesis addresses this need by focusing on extracting and analysing data from a publicly accessible blockchain, specifically Bitcoin, the most prominent cryptocurrency in blockchain.

Cryptocurrencies have initiated a transformative revolution in the global financial landscape. These digital assets, underpinned by blockchain technology, offer unprecedented decentralization, security, and financial autonomy. However, the attributes that make cryptocurrencies appealing, such as pseudonymity and the absence of a central authority, have also made them an attractive playground for malicious actors engaged in fraudulent activities, money laundering, and other illicit endeavours. Therefore, ensuring the integrity and security of cryptocurrency networks has become a paramount concern.

In this context, detecting anomalies in cryptocurrency transactions is a critical challenge. Anomalies can encompass a wide range of activities, from suspiciously large transactions to the utilization of complex transaction patterns, making them difficult to pinpoint using traditional methods. The dynamic and evolving nature of cryptocurrency networks further complicates the task of anomaly detection, requiring innovative and adaptable approaches.

This thesis explores the development of an anomaly detection system for Bitcoin transactions, one of the most widely adopted cryptocurrencies. Leveraging the power of machine learning, we delve into the vast sea of Bitcoin transaction data contained within its blockchain. The objective is to create a system capable of identifying and categorizing anomalies within Bitcoin transaction blocks, thereby safeguarding the security and trustworthiness of the network. Farren, Pham, and Alban-Hidalgo (2016) illustrated that fraud detection within various transactions conducted on blockchain systems is the core issue under investigation. This problem extends beyond financial contexts and can be extrapolated to encompass other blockchain networks, including those in the healthcare sector, public administration, etc. Consequently, this thesis delves into the broader challenge of anomaly detection within blockchain technology, addressing its applicability across a spectrum of blockchain use cases.

2 CHAPTER 2: LITERATURE REVIEW

Anomaly detection finds extensive application in diverse fields, including fault detection, intrusion detection, fraud detection, and various other domains. This thesis predominantly centres on the identification of suspicious transaction blocks. Anomaly detection, a well-established field, encompasses multiple sectors such as credit card fraud, virus intrusion, insurance fraud, and more. Researchers in these domains commonly employ a range of methodologies, including machine learning algorithms and network analysis techniques, to conduct their analyses.

Recent developments in the industry and previous research, such as the work conducted by Bolton, Hand et al. (2001), indicate that unsupervised machine learning algorithms exhibit a more targeted approach to detecting anomalies. On the other hand, research by Phua et al. (2010) suggests that most fraud detection studies rely on supervised machine-learning techniques. These studies emphasize the creation of complex models that can learn and recognize the patterns of anomalies within training data. The choice between supervised and unsupervised methods often depends on a given application's specific requirements and goals, highlighting the need for flexibility and adaptability in fraud detection. This chapter provides an in-depth exploration of how previous research efforts have addressed the challenge of anomaly detection within cryptocurrency transactions. It offers a comprehensive review of the various methodologies and strategies employed in previous studies, along with the outcomes and insights they have derived. This chapter aims to establish a foundation for the novel approaches and contributions presented in this thesis by synthesizing the findings from these prior investigations. It sheds light on the evolution of anomaly detection techniques within cryptocurrency transactions, providing a valuable context for the research in subsequent chapters.

V. Patel, L. Pan, and S. Rajasegarar (2020) focus on anomaly detection in the Ethereum blockchain, which supports smart contracts and various transaction activities. The proposed framework combines graph representation and neural networks to detect anomalies effectively. It models Ethereum data using a graph structure and utilizes a one-class graph neural network for accurate anomaly detection. This research introduces a novel approach to Ethereum anomaly detection, leveraging graph-based techniques to enhance network security and integrity.

T. Pham S. Lee (2016) addresses anomaly detection in network structures, focusing on financial transactional networks where anomalies can involve fraudulent activities. They employ three unsupervised learning methods, k-means clustering, Mahalanobis distance-based approach, and support vector machines, to analyse two graphs derived from the Bitcoin network, one representing the user and the other transactions.

B. Podgorelec, M. Turkanović, and S. Karakatič (2019) present a novel approach to simplify the digital signing process for blockchain transactions, addressing the complexity and security concerns associated with mutual signing. The proposed method includes a personalized anomaly detection system based on user-specific transaction data. An automated signing process, personalized anomaly detection, and user data management in a local environment are the essential contributions of this research.

K. Martin, M. Rahouti, M. Ayyash, and I. Alsmadi (2021) offer a comprehensive digital currency landscape perspective by combining three key elements: blockchain technology, network representation, and unsupervised machine learning. It utilizes raw data collected from digital currency networks, treating them as graphs and expressing these networks as adjacency structures to facilitate analysis. The research examines networks of different sizes and complexities, using network representation learning for transactions and user information. It evaluates a range of metrics to gauge model performance and dataset suitability for these approaches, going beyond simple accuracy to gain deeper insights into the results.

J. Kim, M. Nakashima, W. Fan, and S. Wuthier (2022) present an anomaly detection system on semi-supervised learning that can detect previously unseen patterns through one-class learning. The detection engine is built on AutoEncoder (AE) for profiling background patterns. The study evaluates the security mechanism using actual and simulated data, assessing the impact of various features, such as packet counting and bandwidth, on detection performance and time complexity. A comparison is made with other machine learning models, demonstrating the effectiveness of the AE-based security mechanism with real-time analysis capability. The work is designed to be generally applicable across blockchain applications that use P2P networking, with different model references based on training data specific to each application.

Li, J., Gu, C., Wei, F., Chen, X (2020) delve into blockchain anomaly detection, acknowledging the complexities and challenges of applying detection technology to blockchain systems. These anomaly detection methods mainly exploit four kinds of data mining techniques: statistical (e.g., SVD), conventional machine learning (e.g., k-means and RF), deep learning (e.g., autoencoder), and graph learning (e.g., graph embedding). They suggest that the choice of method should depend on the task's complexity and the dataset's size. Statistical techniques and conventional machine learning models suit simple tasks or smaller datasets. Deep learning and graph learning models are preferred for complex tasks with ample data.

P. Monamo, Marivate, and Bheki Twala (2016) approached the issue of Bitcoin fraud detection with a slightly different method. They utilized a Bitcoin dataset provided by Brugere (2013) from the Laboratory for Computational Biology at the University of Illinois, which, unfortunately, is no longer accessible for unforeseen reasons. Their research's objective was to employ K-means clustering and a variation of K-means proposed by Cuesta-Albertos, Gordaliza, Matran, et al. (1997) in a multivariate context to identify suspicious users and their associated anomalous transactional activities. According to the authors, K-means clustering effectively groups similar instances but cannot detect anomalies. On the other hand, the Local Outlier Factor (LOF) is a popular method for outlier detection, but it is challenging to scale for large datasets due to its computational complexity. This paper aimed to propose an approach that could address the limitations of these algorithms. The results of this study showed a significant improvement, as it successfully detected five fraudulent users out of a total of 30, demonstrating its effectiveness.

J. Lorenz, M. I. Silva, D. Aparício, J. T. Ascensão, and P. Bizarro (2021) exhibit that a lack of labeled data makes supervised ML methods challenging. This study focuses on detecting money laundering with limited labels and shows that it is impossible to see illicit transactions without labels. With just a tiny percentage of labelled data, active learning methods can achieve results similar to supervised methods. This research addresses the challenges of detecting money laundering in a dataset with limited labelled examples, bridging the gap between unsupervised and supervised AML methods, using a real-world dataset with many positive cases and active learning in the cryptocurrency context.

N. Pocher, M. Zichichi, F. Merizzi, MZ. Shafiq and S. Ferretti (2023) acknowledge the necessity of user anonymity and identifiability in the financial sector. It discusses the increasing interest from law enforcement and private sector RegTech providers in tracking transactions across blockchain ecosystems. They apply machine learning techniques, network analysis, and transaction graph analysis to enhance AML efforts. They present experiments using Bitcoin transaction data represented as a directed graph network and highlight the potential of Graph Convolutional Networks (GNC) and Graph Attention Networks (GAT) for AML. The results indicate that these graph-based neural network approaches outperform traditional supervised learning methods.

Z. Gu, D. Lin, and J. Wu (2022) highlights the risks associated with centralized exchanges, including internal operations, business ethics, and asset theft. The paper focuses on the scenario of centralized exchanges and their susceptibility to fraudulent activities such as market manipulation and pump-and-dump schemes. It collects transaction records from significant trades and analyses the impact of different transaction information on transaction amounts. It identifies core factors affecting transaction amounts and proposes an anomaly detection framework based on deep learning models. The framework is designed to detect abnormal transaction amounts and provide early warning for regulatory authorities and investors. The paper's contribution includes the creation of a dataset for exchange transaction data, the development of an anomaly detection framework, and the potential for predicting short-term transaction amounts to assist investors.

Huang et al. (2017) address the problem of blockchain network anomaly detection by focusing on network behavioural patterns. They introduce the behaviour pattern clustering (BPC) technique to categorize behavioural patterns within blockchain data. Their approach involves extracting sequences of changing transaction amounts over time and clustering them into categories. To measure the similarity between lines, the tested various algorithms such as Euclidean distance, Morse and Patel (2007) Dynamic Time Warping (DTW), Chen, Özsu, and Oria (2005) Edit Distance on Real sequence (EDR), and Longest Common Sub-Sequences (LCSS) by Vlachos, Kollios, and Gunopulos (2002). DTW was selected as the sequence measure due to its effectiveness in handling noisy blockchain data. The authors introduce a customized version of K-means called Behaviour Pattern Clustering (BPC) and compare its result with other methods like the Hierarchical Clustering Method (HIC) and Density Based Method (DBSCAN). Their study found that the BPC algorithm was more effective than existing methods, offering an algorithmic approach to blockchain network anomaly detection.

N. Kumar, A. Singh, A. Handa and SK. Shukla (2020) makes several contributions to the field of blockchain security. It involves collecting and categorizing malicious Ethereum addresses, spanning various attack types. Additionally, non-malicious addresses are labelled through meticulous data preprocessing. The study undertakes feature extraction and engineering to identify relevant attributes for adequate classification. Notably, the research successfully detects malicious nodes within the Ethereum network with high accuracy.

3 CHAPTER 3: BACKGROUND

Throughout the history of finance, using ledgers has been a practice as old as the concept of currency itself. It dates back to ancient times, reminiscent of the challenges faced during the Roman Empire, where achieving consensus in a decentralized and trust less manner was essential. This dilemma aligns with the Byzantine General's problem, which illustrates a scenario where a dispersed army requires coordinated action to succeed, but miscommunication can lead to failure. Bitcoin's innovative solution to the Byzantine General's problem lies in introducing proof of work, a unified protocol that addresses the synchronization and trust issues. This concept forms the bedrock of distributed ledger technology, emphasizing the need for decentralized, synchronized maintenance of the ledger by all participants, revolutionizing the world of blockchain and cryptocurrencies.

In recent years, distributed ledger technology, commonly exemplified by blockchain, has redefined the landscape of financial, specifically cryptocurrency transactions, with Bitcoin being a prominent example. The blockchain serves as a decentralized, tamper-resistant digital ledger where transactions are recorded in a sequential chain of blocks, each linked to the preceding one through cryptographic hashes. This groundbreaking technology eliminates the need for central authorities, relying instead on a network of nodes to reach consensus on the validity of transactions, offering attributes such as decentralization, transparency, immutability, and enhanced security. These features have revolutionized cryptocurrencies and found applications in various industries. These ledgers have taken multiple forms, from clay tablets and stones to papyrus scrolls and paper documents. With the advent of the digital age, ledgers transitioned into spreadsheets. While early digital ledger systems often imitated traditional double-entry bookkeeping methods, they struggled to handle the vast amount and geographic spread of data. The emergence of distributed ledgers was made possible by advancements in computational power, cryptography, and sophisticated algorithms addressing these limitations.

3.1 DISTRIBUTED LEDGER TECHNOLOGY

Distributed ledger technology (DLT) is a digital system that records asset transactions, where transaction details are simultaneously stored in multiple locations. Unlike traditional databases, DLT operates without a central data repository or administrative control. It encompasses the technological infrastructure and protocols that enable the concurrent access, validation, and updating of records. This technology functions across a network of nodes or entities, where each node processes and verifies every transaction, creating a consensus on their accuracy. DLT can record both static data, such as registries, and dynamic data, like financial transactions. Blockchain, driven by Bitcoin, was a pioneering example of DLT, which gained significant interest after Bitcoin's launch in 2009. Various industries, including finance, healthcare, and supply chain management, have since explored DLT applications to streamline processes.

Historically, organizations have maintained data across various locations, either on paper or within isolated software systems, centralizing the data intermittently. For instance, different divisions within a company might hold their data and contribute it to a central ledger only when necessary. Likewise, collaborative entities typically manage their own data and provide it to a centralized ledger under authorized control as needed. DLT's key advancement lies in its ability to reduce or eliminate the time-consuming and error-prone reconciliation process associated with these disparate contributions, ensuring that all participants can access the most current, trustworthy version of the ledger.

Sunyaev A. (2020) narrates that Distributed Ledger Technology (DLT) has garnered significant attention in recent years and is considered one of the most talked-about technologies. The widespread adoption of cryptocurrencies, particularly Bitcoin, has made blockchain and DLT a prominent subject in both practical applications and research. Cryptocurrencies are digital assets designed for use as a medium of exchange, utilizing cryptographic techniques to ensure the security and integrity of transactions and represented assets.

DLT is founded on the principle of decentralization, distinguishing it from traditional centralized databases. It operates within a peer-to-peer (P2P) network where multiple nides collaborate to store, validate, and update the ledger concurrently. This architecture eliminates the necessity for a central authority, significantly reducing the risk associated with a single point of failure. The process initiates with the replication of digital data across the network, where each node maintains an identical ledger copy and independently process new transactions. The consensus algorithm ensures agreement among all participating nodes on the correct ledger version, and once consensus is achieved, the updated ledger is synchronized across all nodes, preserving accuracy. DLT employs cryptography for secure data storage and utilizes cryptographic signatures and keys to permit access solely to authorized users. The technology creates an immutable database, ensuring that stored information cannot be deleted, and all updates are permanently recorded.

This architectural shift revolutionizes how information is collected and shared, as it moves away from a centralized authoritative location to a decentralized system where all relevant entities have access to and can modify the ledger. This enhanced transparency in DLT fosters a high level of trust among participants and substantially reduces the likelihood of fraudulent activities within the ledger. Consequently, DLT eliminates the reliance on trusted central authorities or third-party providers, providing a robust defence against manipulation.

Distributed Ledger Technology (DLT) presents a transformative shift in record-keeping by departing from the traditional centralized control found in paper-based and conventional electronic ledgers. Centralized systems demand significant labour and computational resources to maintain control, often resulting in incomplete or outdated ledgers vulnerable to errors and manipulation from various contributions. Verification of data accuracy among participants is inefficient in such setups. In contrast, DLT introduces real-time data sharing, ensuring an always up-to-date ledger with enhanced transparency. It inherently enhances security by eliminating a single point of failure and target for cyberattacks, common in centralized ledgers. While DLT has the potential to expedite transactions and reduce costs, the decentralized verification process and ledger distribution require substantial computational resources, which can affect performance in specific network environments compared to centralized ledgers.

3.1.1 BLOCKCHAIN TECHNOLOGY

Blockchain is a highly secure method of recording and storing information through a distributed ledger. It operates by duplicating and distributing transaction records across a network of connected computers, creating an immutable chain of blocks. Each transaction is authenticated by the owner’s digital signature, ensuring its protection against tampering. It is akin to a shared google spreadsheet among networked computers, providing transparency while safeguarding data integrity.

Blockchain technology, often deemed as not hackable, is susceptible to 51% attacks, allowing malicious actors to control over 50% of a blockchain’s computational power and manipulate the ledger’s integrity. This reality underscores the need to view blockchain as a valuable technology rather than a magical solution. The 51% attack exploits the 51% problem, where a single party with majority mining power can forge entries, leading to double spending and potential chain forking in their favour.

Two main types of blockchains exist: public and private. Public blockchains use the open internet for transaction validation and ledger updates, potentially posing confidentiality concerns for enterprises. In contrast, private blockchains are more restricted, allowing known organizations to participate. They emphasize participant identity, with public blockchains favouring anonymity and private ones employing selective endorsement by recognized users for secure, permissioned networks. Ensuring security in rapidly evolving blockchain technologies requires developers to prioritize securing applications and services. This involves activities such as risk assessments, threat modelling, and code analysis, incorporating security measures from the outset to achieve successful and safe blockchain applications.

Blockchain gets its name from the way it stores transaction data in blocks linked together to form an unchangeable chain. As more transactions occur, the blockchain grows, with each block recording and validating transaction details according to network rules agrees upon by its participants. Every block consists of a unique identifier (hash), a timestamp for recent valid transactions, and the previous block’s hash, preventing alteration or insertion of blocks. This design theoretically ensures the blockchain’s security.

Four fundamental blockchain concepts include a shared ledger, permissions, smart contracts, and consensus. A shared ledger serves as an append-only distributed record system shared among network participants, reducing redundant data entry. Permissions guarantee secure, authenticated, and verifiable transactions, aiding organizations in regulatory compliance. Smart contracts are blockchain -stored agreements or rules governing business transactions, automatically executed during transactions. Consensus mechanisms, such as proof of stake and multi signature, ensure that all parties agree on verified network transactions. Blockchain networks involve various participants, including blockchain users with transaction permissions, regulators overseeing network transactions, blockchain network operators managing network aspects, and certificate authorities issuing and managing the required certificates for permissioned blockchains.

3.1.2 HASHGRAPH

Hashgraph is an innovative form of distributed ledger technology (DLT) developed by Leemon Baird, utilizing a directed acyclic graph (DAG) data structure to establish a consensus protocol for data validity among distributed systems. It addresses the challenge of Byzantine failures through a combination of a gossip protocol and virtual voting. Unlike traditional gossip, Hashgraph spreads transaction information and their recipients, ensuring anonymity regarding the transaction's creator while preventing forgery or the spread of fake transactions. This unique approach enhances security and trust in the network, as no one can determine the transaction's origin or authenticity when shared second-hand.

In addition, Hashgraph consensus represents an advancement beyond traditional blockchain consensus mechanism. Unlike blockchains, which rely on the computational power of vast networks to validate transactions, Hashgraph employs a protocol based on node communication for recording and confirming transactions. While both Hashgraph and blockchain are decentralized ledgers secured through its consensus process. It achieves consensus using "gossip", "gossip about gossip" and virtual voting, addressing common issues associated with consensus-building algorithms like proof of work (POW). This innovative approach enhances speed and efficiency, making Hashgraph a promising alternative to traditional blockchain consensus methods.

Hashgraph and blockchain are both technologies designed to create secure, decentralized ledgers for data storage. While blockchain emerged in 2008 as the foundation for Bitcoin, Hashgraph is a more recent innovation from 2016. Hashgraph boasts several advantages over blockchain, such as faster transaction speeds, increased throughput capacity, and advanced consensus algorithms. While blockchain was originally created to establish a secure, distributed ledger for Bitcoin transactions, Hashgraph's consensus mechanism enables a broader range of applications beyond just cryptocurrency, including smart contracts and various distributed uses.

The choice between Hashgraph and blockchain as technologies depends on the specific application's requirements. In cases where faster transaction speeds and increased throughput are crucial, Hashgraph may be the more suitable choice. On the other hand, if an application demands a strong emphasis on security and decentralization, blockchain might be the preferable option. Ultimately, developers have the flexibility to determine which technology aligns better with their particular project needs. Both Hashgraph and blockchain are valuable tools for distributed applications, each with its own strengths.

3.2 BLOCKCHAIN NETWORK

Blockchain serves as an immutable ledger that enhances the ease of recording transactions and managing and managing assets in corporate networks, whether tangible or intangible. It offers accessibility and security for recording and trading a wide array of valuable assets, ultimately reducing risk and cutting costs for all involved parties. A blockchain network, the technical infrastructure facilitating these capabilities, enables applications to access ledger and smart contract services. Smart contracts are central for initiating transactions, which are then securely and unalterably recorded on each peer node's copy of the ledger. App users, including end-users and administrators, interact within this network, allowing for tracking various aspects like orders, accounts, payments and production. The transparency of blockchain networks ensures that all transaction details are shared among members, instilling confidence and fostering efficiency and opportunities. These networks typically emerge from consortiums formed by multiple organizations, governed by collectively agreed-upon policies. Blockchain networks can further be categorized as public, private, or permissioned, each serving distinct purposes and audiences.

Blockchain technology offers numerous advantages, yet its vulnerability to attacks and centralization in networks with an unstable ecosystem or an unverified consensus process is a concern. Striking a balance among scalability, decentralization, and security, known as the Blockchain Trilemma, is a key challenge. Environmental issues are also in the in the spotlight, as proof-of-work (POW) consensus consumes substantial electricity. Additionally, blockchain's technical complexity and its perceived complexity may intimidate businesses and individuals.

The rapid ascent of cryptocurrencies marked just the initial stage of blockchain's integrations into various sectors and daily life. An increasing number of industries are exploring blockchain's potential, while more people recognize its utility in everyday goods and services. As the blockchain industry continues to expand, it shows no signs of slowing down and may even evolve to become an integral part of our digital architecture in the future, potentially reshaping the digital landscape.

3.2.1 SCOPES OF BLOCKCHAIN

The future of blockchain technology holds immense promise across various industries. With its disruptive potential, it has captured the attention of organizations in diverse fields. Financial institutions have invested in blockchain due to its effectiveness in tracking financial assets and its potential to combat illicit financial activities. Governments are also exploring its application to enhance financial regulations. Furthermore, blockchain's future lies significantly in cybersecurity, offering secure and verifiable data storage, reducing vulnerabilities, and bolstering cloud security. It is increasingly being integrated into Internet of Things (IoT) networks and supply chains, improving transparency, reliability, and efficiency. The rise of cryptocurrencies, like Bitcoin, suggests a potential future where governments may develop their digital currencies. Finally, blockchain is transforming data management for government entities, enhancing operational efficiency.

Blockchain technology has given rise to numerous innovative applications such as cryptocurrencies, NFTs, DeFi, and more. These advancements are shaping a future where asset exchanges across various categories are becoming increasingly prevalent. Many companies are actively developing blockchain solutions, and the emergence of the Metaverse is expected to drive these technologies forward, potentially revolutionizing various aspects of human society. However, this rapid development also brings significant risks that must be carefully managed.

In recent years, the investment landscape has expanded significantly due to these technologies. Traditional investment options, like the stock market, fixed deposits, and gold, now share the stage with blockchain-based opportunities. If harnessed in the right direction, this diversification of investment avenues has the potential to stimulate economic growth. As of 2022, there is a growing demand for Blockchain developers, and more individuals are entering the Blockchain community. The proliferation of cryptocurrency exchange platforms has piqued the interest of many people, leading to a surge in curiosity and a desire to gain knowledge about these technologies. This trend reflects the increasing importance and relevance of blockchain in this modern world.

3.2.2 CONSENSUS OF BLOCKCHAIN

In the realm of blockchain technology, a consensus mechanism plays a pivotal role by validating and authenticating transactions, making them trustworthy within the network. Prominent cryptocurrencies like Bitcoin and Ethereum rely on these mechanisms to bolster security and maintain the integrity of their blockchain systems. Essentially, consensus mechanisms serve as a foundation of trust in the world of cryptocurrency trading, ensuring that legitimate transactions are faithfully recorded and preventing potentially fraudulent activities. These mechanisms expedite the inclusion of validated transactions in the blockchain, an imperative aspect of decentralized trading. To accomplish this, various methodologies are employed to foster security, trust, and agreement across the blockchain network, ultimately reinforcing the credibility of the system and its transactions.

Proof of Work (POW)

Bitcoin's Proof of Work (POW) consensus mechanism ensures blockchain security through a competitive mining process that relies on solving complex cryptographic puzzles, which consumes substantial computational power and energy. The miner who successfully validates a block receives newly created bitcoins and a transaction fee. Despite being a foundational technology, POW faces criticism due to its energy consumption, scalability challenges, and concerns over centralization. Centralization arises from the dominance of miners with access to powerful hardware, and there is a of collusion among large mining pools. As a result, the blockchain community continues to explore alternative consensus algorithms such as Proof of Stake (POS) and Proof of Activity to address these issues and improve blockchain technology's efficiency, decentralization, and sustainability.

Proof of Stake (POS)

Proof of Stake (POS) is a consensus algorithm that contrasts with Proof of Work (POW) by eliminating mining and replacing miners with validators. Validators invest their cryptocurrency stakes and compete to validate blocks. They are selected to confirm new transactions based on their proportional stake in the network. When validator proposes a block that is added to the chain, they receive rewards proportional to their stake. POS makes the mining process virtual and resource-efficient, as it depends on coin ownership rather than computational power. While POS offers a more environment friendly and decentralized alternative to POW, it faces the "nothing-at-stake" problem, where validators could potentially double-sign and fork the system. Some solutions involve locking currency in virtual vaults and making modifications to maintain a broader and more secure network. Ethereum is planning a transition to POS, leveraging the Casper protocol, while there is a variation called Delegated Proof of Stake (DPOS), where individuals delegate their stake representation to overarching entities, allowing smaller stakeholders to collaborate for better representation. However, DPOS introduces greater network centralization.

Proof of Activity (POA)

Proof of Activity is an alternative incentive structure for Bitcoin that combines both Proof of Work (POW) and Proof of Stake (POS) in a hybrid approach. POA starts with traditional POW mining, where miners compete to solve a cryptographic puzzle, but the blocks mined only serve as templates and do not contain transactions. The system then switches to POS, selecting a random group of validators to sign the block based on the block's information. Validators with more coins have a higher chance of being chosen. The template becomes a full block when all validators sign it, and fees are shared between the miners and validators. POA faces criticisms similar to POW (high energy consumption) and POS (potential double signing by validators). Currently, Decred is the only coin utilizing a variation of POA.

Practical Byzantine Fault Tolerance Algorithm (PBFT)

The Practical Byzantine Fault Tolerance Algorithm (PBFT) was developed as a solution to the Byzantine Generals Problem (BGP) in the context of distributed networks and blockchains. In this allegory, the "generals" represent participants in a blockchain network, and the "loyal generals" aim to collectively decide the validity of information submitted to the blockchain. PBFT is one of several solutions to the Byzantine general's problem and is employed by blockchains like Hyperledger, Stellar, and Ripple. The algorithm operates by having each general maintain an internal state and make decisions about incoming messages based on this state. These decisions are then shared with other generals, and a consensus is reached based on the total decisions made by all participants. PBFT offers a lower-effort consensus mechanism suitable for low-latency storage systems and digital asset platforms with a high volume of transactions.

Hyperledger's approach for consensus

Hyperledger, utilizing the Practical Byzantine Fault Tolerance Algorithm (PBFT), enables the creation of digital assets in a distributed ledger network and explores broader applications beyond asset-based systems. PBFT ensures that corruption issues are contained, and identify verification, access control, and ongoing checks are integrated throughout the transaction flow. This approach goes beyond transaction order and encompasses the entire process, from proposal to commitment, providing data integrity protection and enabling the implementation of various consensus algorithms. Permissioned networks like Hyperledger Fabric achieve higher transaction throughput rates and improved performance by replacing resource-intensive consensus methods with more efficient ones.

3.3 BLOCKCHAIN SECURITY

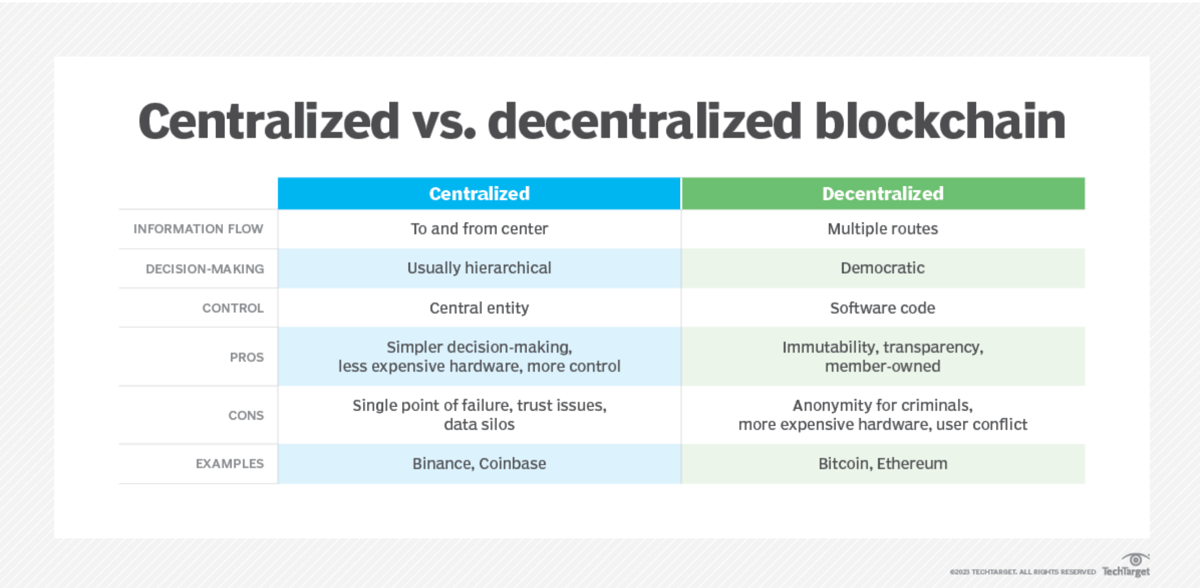
Blockchain is renowned for its security due to decentralized control and cryptographic hashing. With no central authority to target, hacking all network nodes is exceptionally challenging, making tempering with the Blockchain difficult. Cryptographic hashing ensures data integrity by linking each block securely. However, no system is entirely immune to threats, emphasizing the need for basic security measures to safeguard data.

A Blockchain is a decentralized digital ledger managed by a network of computers, ensuring transparency and security through shared verification. Unlike traditional central authority-ledgers, all parties on the Blockchain participate in recording and verifying transactions using intricate mathematical algorithms. Once verified, transactions are permanent and tamper-proof. This decentralized feature is ideal for applications demanding heightened security and transparency, such as financial transactions and supply chain management. Blockchains are considered a groundbreaking technology with the potential to revolutionize various industries and transform our daily tech interactions, evidenced by the growing interest in Blockchain security jobs and projects.

Blockchain technology offers enhanced transaction security through cryptography, decentralization, and consensus, with the global blockchain market projected to reach $20 billion by 2024. Many banks, approximately 69%, are actively exploring blockchain to bolster the safety, consistency, and simplicity of their services. Recent cyberattacks within the blockchain space, such as the DAO code exploitation attack and the Bithumb crypto exchange breach, highlight security vulnerabilities. Organizations worldwide are harnessing blockchain for distributed databases, digital transactions, cybersecurity, and healthcare solutions. While it brings substantial benefits, the technology’s adoption has also attracted cybercriminals seeking to exploit and target organizations through cyberattacks.

3.3.1 DECENTRALIZATION

Centralized and decentralized structures represent opposing paradigms. Centralization involves control vested in a central authority or entity, such as a nation's central bank overseeing its currency. In contrast, decentralization signifies a lack of singular ownership, management, or control within a network or structure. While some cryptocurrencies adhere to decentralization, not all do. Prominent examples like Bitcoin and Ethereum's Ether follow decentralized principles, relying on code and community-driven monetary policies rather than central bank regulation. Bitcoin's peer-to-peer public blockchain employs a cryptographic protocol called proof of work (POW) to validate and add blocks of transaction data. This POW-based system allows users to contribute to the blockchain, ensuring transparency and accessibility as the blockchain is publicly viewable and open for block addition through POW-verified transactions.



Centralized vs. decentralized blockchain

	Centralized	Decentralized
INFORMATION FLOW	To and from center	Multiple routes
DECISION-MAKING	Usually hierarchical	Democratic
CONTROL	Central entity	Software code
PROS	Simpler decision-making, less expensive hardware, more control	Immutability, transparency, member-owned
CONS	Single point of failure, trust issues, data silos	Anonymity for criminals, more expensive hardware, user conflict
EXAMPLES	Binance, Coinbase	Bitcoin, Ethereum

Figure 01: Differentiation between Centralized & Decentralized Blockchain

(Source: techtarget.com)

The primary objective of decentralization in blockchains is to prevent concentration of control in the hands of a few entities or central banks, aligning with the original idea of cryptocurrencies for peer-to-peer transactions, bypassing traditional banking systems. Decentralized blockchains ensure immutability, making data entries irreversible once recorded, and allowing the addition of new data without altering the old. This immutability is exemplified by Bitcoin, where transactions are permanently visible to all, akin to an advanced form of eBay feedback. Notably, not all digital currencies follow this decentralized model, as some employ private, centralized systems where select individuals hold the authority to add blocks and validate transactions, often utilized in privacy-focused sectors like healthcare and finance.

Decentralization offers a myriad of advantages, such as built-in trust without the need for prior knowledge or trust in transaction parties, enhanced data accuracy through consistent, interlinked data, reduced downtime by eliminating single points of failure, transparency as decentralized blockchains are publicly accessible, user control over their information, and immutability due to the consensus requirement of all network nides for data alterations. Furthermore, proponents, relying on encryption techniques, whether symmetric or asymmetric, to safeguard data from tampering. Decentralized blockchain come with their share drawbacks, including increased costs due to the need for additional systems and personnel, potential challenges in reaching a consensus when multiple voices have a say, difficulties in achieving clarity in decision-making processes, and a lack of formal discipline and accountability, which can lead to inefficiencies or neglect in network maintenance and performance.

Decentralized blockchain networks, exemplified by popular cryptocurrencies like Bitcoin and Ethereum, are characterized by their absence of corporate ownership and are known for their decentralized, secure, immutable, and permissionless nature. Ethereum, in particular, is renowned for its decentralized nature and ability to facilitate the creation and deployment of smart contracts and decentralized applications (dApps) without the need for intermediaries. However, Ethereum's recent transition from the proof of work (POW) consensus mechanism to proof of stake (POS) has raised concerns about increased centralization, despite its energy-efficient benefits.

3.3.2 BYZANTINE PROBLEM

In 1999, Miguel Castro and Barbara Liskov developed the Practical Byzantine Fault Tolerance (PBFT) algorithm, enabling network nodes to achieve consensus without relying on a central coordinating entity. PBFT's core aim is to provide practical Byzantine state machine replication, ensuring resilience against malicious nodes. The effectiveness of the PBFT model hinges on ensuring that the number of malicious nodes does not exceed one-third of the total nodes within a vulnerability window, which becomes less likely as the network grows larger. However, its practical applications were limited due to the exponentially increasing time required to reach a consensus as the network expands. In 2008, Byzantine Fault Tolerance gained significant attention with the release of the Bitcoin white paper by Satoshi Nakamoto, introducing a groundbreaking consensus mechanism resilient to Byzantine faults, utilizing the proof of work (POW) protocol. This innovation has since paved the way for the development of various blockchain consensus methods, including proof of stake (POS), all aimed at achieving Byzantine fault tolerance.

Blockchain technology offers the potential to revolutionize the way digital assets are exchanged and ownership is securely tracked without the need for centralized control. This technology is underpinned by a distributed by a distributed ledger that maintains consistency, even in the pressure of potentially malicious participants or Byzantine failures. Blockchain combines insights from various fields, including distributed computing, cryptography, and game theory. It utilizes a distributed network to ensure ledger data persistence, employs public key cryptography for secure asset transfers, implements incentive mechanisms to validate transactions, and relies on Byzantine fault-tolerant consensus protocols to maintain ledger integrity. The consensus problem in blockchain is about enabling non-faulty processes to agree on the order of transactions in a block, which includes properties like agreement, validity, and termination.

V. Gramoli highlights the risks associated with using proof-of-work blockchains like Bitcoin and Ethereum without a comprehensive understanding of their guarantees. It identifies vulnerabilities affecting these systems and their consensus algorithms. He presents existing attacks on Bitcoin's consensus protocol and discusses potential vulnerabilities in Ethereum, emphasizing the importance of research to design secure consensus algorithms suitable for blockchain applications, especially in scenarios involving volumes of assets, private blockchains, or consortium blockchains.

3.4 CRYPTOCURRENCY

Cryptocurrency is a digital currency created using encryption algorithms and operates as an alternative form of payment, serving both as a currency and a virtual accounting system. To use cryptocurrencies, individuals require a cryptocurrency wallet, which can be cloud-based or stored on a computer or mobile device and holds encryption keys confirming identity and linking to the cryptocurrency. However, using cryptocurrencies involves several risks, including their newness and high market volatility, lack of regulation or insurance, vulnerability to hackling like other digital assets, and the potential for complete loss in the event of losing access to the digital wallet or its backups.

Cryptocurrencies operate on a decentralized public ledger known as the blockchain, serving as a comprehensive record of all transactions maintained by currency holders. The creation of cryptocurrency units involves a process called mining, where complex mathematical problems are solved using computer power to generate coins. Users can also purchase these currencies from brokers and manage them through cryptographic wallets. It is important to note that cryptocurrency ownership entails possessing a digital key enabling the transfer of a record or unit of value between parties without the need for a trusted intermediary. While Bitcoin, introduced in 2009, marked the beginning, the financial applications of cryptocurrencies and blockchain technology are still evolving, with expectations of expanded usage in areas like trading bonds, stocks, and other financial assets.

There exists a multitude of cryptocurrencies, with some of the most renowned including:

Bitcoin: Launched in 2009, Bitcoin was the inaugural cryptocurrency and continues to be the most widely traded. It was created by an individual or group under the pseudonym Satoshi Nakamoto, whose true identity remains undisclosed.

Ethereum: Introduced in 2015, Ethereum is a blockchain platform featuring ots own cryptocurrency, Ether (ETH) or Ethereum. It ranks as the second most popular cryptocurrency after Bitcoin.

Litecoin: Bearing many similarities to Bitcoin, Litecoin has been more proactive in adopting new advancements. These include faster transaction processing and an increased capacity for transactions.

Ripple: Established in 2012, Ripple is a distributed ledger system capable of monitoring various transaction types, extending beyond cryptocurrencies. The company responsible for Ripple has collaborated with various banks and financial institutions.

Cryptocurrencies apart from Bitcoin are collectively referred to as "altcoins" to distinguish them from the pioneering cryptocurrency.

Bitcoin transactions, as well as those of other cryptocurrencies like Ethereum, Dash, or Bitcoin Cash, involve the transfer of digital assets on their respective blockchains. These transactions are essentially recorded on the Blockchain, serving as a public ledger. To execute such transactions, users require cryptocurrency clients, commonly referred to as wallets. These wallet applications are software tools enabling users to manage their digital funds, facilitating the sending and receiving of cryptocurrencies through blockchain transactions.

In a cryptocurrency transaction, several crucial elements come into play:

Inputs: These refer to unspent outputs from prior transactions, verifying the source of the assets used in the transaction. They contain the original receipt address for the bitcoins.

Outputs: Outputs include the recipient's address and the amount being transferred. They may also encompass change or return addresses for any remaining funds, allowing multiple outputs within a single transaction.

Identifier (TXid): Each transaction is uniquely identified within the blockchain through a hash generated from its inputs and outputs.

Transaction Fee: Miners receive a small fee for processing transactions and creating new blocks. This fee is not explicitly associated with any output but is left unallocated, signifying a reward for the miner processing the transaction.

3.5 ANOMALY DETECTION

Anomaly detection plays a pivotal role in the financial sector, where identifying unusual patterns, outliers, or fraudulent activities can make the difference between profit and loss. As financial systems grow increasingly complex, with vast amounts of data generated every second, the need for robust anomaly detection methods is more crucial than ever. This article delves into the significance of anomaly detection in finance, its applications, challenges, and the technologies driving its evolution.

Anomaly detection holds immense importance across various domains. It serves as a critical tool for fraud detection, saving financial institutions billions of dollars annually by identifying irregular spending patterns and unauthorized activities. Furthermore, in risk management, it enables the early identification of unusual fluctuations in financial instruments, allowing swift reactions to minimize losses and seize opportunities. Credit scoring benefits from anomaly detection, helping banks assess credit risks by pinpointing unexpected financial behaviour, while algorithmic trading platforms leverage it to identify market anomalies, facilitating profitable trading opportunities with rapid responses. Anomaly detection in finance is a valuable tool, yet it presents several challenges. Imbalanced data, where anomalies are infrequent compared to normal data, can lead to biased models if not managed properly. Additionally, the ever-evolving nature of anomalies, as fraudsters adapt to new techniques, requires constant evolution in detection systems to keep pace. Lastly, ensuring interpretability in complex machine learning models is essential, particularly in finance, where understanding why an anomaly was detected is crucial for informed decision-making and compliance with regulations.

Anomaly detection in finance is increasingly powered by cutting-edge technologies. Machine learning models like Isolation Forests, One-Class SVM, and Autoencoders have demonstrated effectiveness in learning and adapting to evolving data distributions. Real-time data processing capabilities offered by technologies like Apache Spark and Kafka are essential for handling vast amounts of high-frequency trading data and transaction records. Moreover, Natural Language Processing (NLP) plays a role in analysing unstructured data, such as news articles and social media, to identify anomalies in financial sentiment that can influence market dynamics.

3.6 TRANSACTION FRAUD AS AN ISSUE

Transaction fraud encompasses a range of deceptive practices aimed at exploiting payment mechanisms. It occurs through various channels, including credit card fraud, account takeovers, identity theft, and fraudulent online purchases. Common tactics employed by fraudsters include phishing, data breaches, and hacking, often resulting in unauthorized access to sensitive financial information.

Cryptocurrency transaction fraud: It is a growing concern within the digital asset space. It encompasses a range of deceptive activities aimed at cryptocurrency users and platforms. Phishing attacks, where fraudsters trick individuals into revealing private keys or access credentials, are a common form of such fraud. Investors have fallen victim to Ponzi schemes and fraudulent Initial Coin Offerings (ICOs) that promise high returns but vanish with the funds. Malware and ransomware are also being used to hijack cryptocurrency wallets or extort ransoms in digital currency. Safeguarding against these evolving tactics necessitates vigilance, robust security measures, and increased user education.

Credit Card Fraud: This involves unauthorized use of credit card information to make purchases or withdraw funds. Fraudsters obtain card details through various means, such as skimming devices, carding, or hacking into online databases.

Account Takeovers: In an account takeover, criminals gain unauthorized access to a person's or business's financial accounts. This can lead to unauthorized transactions, changes in account information, and other malicious activities.

Identity Theft: Identity theft occurs when a fraudster assumes another person's identity to conduct fraudulent financial transactions. This may involve opening credit accounts, taking out loans, or making unauthorized purchases.

Online Purchase Fraud: Fraudulent online transactions can be perpetrated through various schemes, including fake e-commerce websites, stolen credentials, or spoofed payment gateways.

3.6.1 FINANCIAL FRAUDS

Financial transaction fraud is a pervasive and evolving threat that poses significant risks to individuals, businesses, and financial institutions worldwide. In June 2016, former JP Morgan Chase Vice President Hernán Arbizu was arrested by the Argentine Federal Police for a series of fraudulent activities. Arbizu, who had worked at several prestigious banks throughout his career, secretly managed various bank accounts from his previous employers, carrying out unauthorized transfers and laundering money for some of his clients. He later used confidential information to report JP Morgan Chase for tax evasion, but this backfired as he was ultimately extradited to the United States and accused of fraud, money laundering, identity theft, and fraudulent transfers. The case highlights a lack of oversight and poor financial risk management within the bank, allowing Arbizu to exploit the institution's resources for his own illicit activities.

Another example was The Société Générale Bank, they experienced a significant financial fraud due to unauthorized transactions by one of its employees, Jérôme Kerviel, highlighting the issue of rogue traders who independently and recklessly make high-risk investments within financial institutions, potentially causing substantial losses or profits. This case, similar to the Barings Bank and Nick Leeson incident, underscores the need for robust financial risk management to mitigate risks stemming from market movements, economic factors, credit, and liquidity. A company's financial risk analysis should align with its risk acceptance policy, setting clear risk appetite and tolerance levels. Inadequate controls or neglect of financial risks can lead to severe financial losses.

To effectively combat financial fraud, organizations must adopt a comprehensive and tailored approach. Key strategies include conducting regular Enterprise-Wide Risk Assessments (EWRAs) to identify and prioritize unique fraud risks, strengthening internal controls to prevent both internal and external fraud, fostering a culture of fraud prevention through employee training and sound governance, implementing robust cybersecurity measures, and establishing a clear response plan for incident management. This multifaceted strategy is essential to mitigate risks, protect sensitive data, and respond effectively when fraud incidents occur. Each organization must adapt these strategies to its specific risks and operations, ensuring a proactive and holistic approach to fraud prevention.

3.6.2 BLOCKCHAIN & CRYPTOCURRENCY SCAMS

Cryptocurrency has become a hotbed for scams, with numerous incidents reported, including a $320 million loss by the Wormhole exchange due to a cyberattack and over $1 billion stolen by cryptocurrency scammers in 2021, as per the Federal Trade Commission. Unlike traditional currency, cryptocurrency resides in digital wallets and relies on blockchain technology, making it less recoverable if stolen. These scams often employ classic fraudulent tactics, despite the innovative nature of cryptocurrencies, creating a need for increased vigilance among users to protect their digital assets.

Cryptocurrency investment scams come in various forms. One common scheme involves fraudsters posing as experienced "investment managers," promising substantial profits in the cryptocurrency market, but after receiving upfront fees, they simply abscond with the money. In some cases, they request personal identification details, further compromising victims. Another prevalent tactic is using fake celebrity endorsements, where scammers create bogus accounts or articles that suggest celebrities are endorsing lucrative investments, luring individuals with fake claims of financial gains while maintaining the illusion of credibility through reputable brand names like ABC or CBS. In reality, these endorsements are entirely fabricated. Vigilance is essential in the cryptocurrency investment landscape to avoid falling victim to such scams.

Another form of scam is known as Ponzi schemes. It operates in the cryptocurrency realm by using funds from new investors to pay returns to earlier participants, creating a cycle with no legitimate investments. These schemes entice new investors with the promise of substantial profits and minimal risk, but in reality, there are always inherent risks, and guaranteed returns do not exist within this fraudulent framework.

In addition, Phishing scams, a longstanding and persistent threat, continue to be prevalent. Scammers employ tactics like sending deceptive emails containing malicious links to counterfeit websites, aiming to collect sensitive personal information, notably private keys for cryptocurrency wallets. Unlike passwords, private keys are unique and irreplaceable, making their theft a significant concern. If a private key is compromised, users must create an entirely new wallet to update it. To guard against phishing scams, it's essential never to enter sensitive information through email links and, instead, access websites directly, regardless of how authentic the email or link may seem.

4 CHAPTER 4: MACHINE LEARNING AND ANOMALY DETECTION

Machine learning is indispensable for anomaly detection across various domains, including cybersecurity, finance, healthcare, and industrial monitoring. It leverages historical data to autonomously identify deviations, making it a valuable tool for risk mitigation. Machine learning excels at spotting subtle anomalies, adapts to evolving patterns, and efficiently processes large datasets. Its rule-free, data-driven approach ensures operational stability and the effective handling of abnormal events.

Machine learning is integral to anomaly detection for several compelling reasons:

Data Complexity and Volume: With the exponential growth in data volume and complexity, traditional rule-based or statistical methods often fall short in efficiently identifying anomalies. Machine learning algorithms excel at processing vast amounts of data and automatically recognizing patterns.

Handling Unlabelled Data: When anomalies aren't explicitly labelled or defined, it becomes challenging to establish precise rules or thresholds for detection. Machine learning algorithms can learn from unlabelled data, uncovering hidden patterns and deviations from normalcy.

Adaptive Learning: Machine learning models exhibit adaptability to new data, continuously improving their ability to spot anomalies. They evolve their understanding of typical behaviours based on incoming information, enabling the detection of emerging or previously unnoticed anomalies.

Feature Extraction: Effective anomaly detection often involves extracting relevant features or attributes from the data. Machine learning techniques can autonomously identify and extract valuable features from complex datasets, enhancing the accuracy and efficacy of anomaly detection.

Detecting Complex Anomalies: Anomalies can manifest in diverse and intricate forms, posing a challenge for conventional techniques. Machine learning, including deep learning models, can discern complex correlations and patterns in data, facilitating the detection of intricate anomalies.

In this thesis, we have considered unsupervised K-means Clustering, supervised Principal Component Analysis and Random Forest to build the anomaly detection model.

4.1 K-MEANS CLUSTERING

The K-means algorithm is an iterative method designed to divide a dataset into K pre-defined, distinct, and non-overlapping clusters. Each data point is allocated to one group, ensuring that intra-cluster data points are as similar as possible while maintaining maximum dissimilarity between clusters. This assignment of data points to clusters aims to minimize the sum of squared distances between data points and the centroid of their respective cluster, where the centroid represents the arithmetic mean of all data points in that cluster. Essentially, the algorithm seeks to minimize variation within clusters, thereby enhancing the similarity or homogeneity of data points within each cluster.

K-Means is a straightforward unsupervised clustering algorithm employed to group data into K clusters. It operates through iterative assignments of data points to one of the K clusters, determined by their proximity to the cluster's centroid. The outcomes of the K-Means algorithm include K cluster centroids and the classification of data points into their respective clusters.

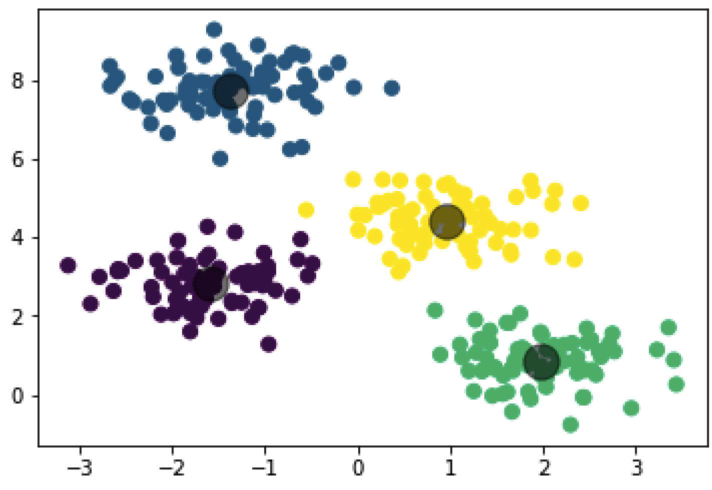


Figure 02: K-Means Clustering – Clustering with centroids marked

(Source: muthu.co)

Algorithm Overview:

Given input data points x1, x2, x3, ..., xn, and the desired number of clusters, K, the algorithm follows these steps:

- ✓ Initialize K centroids by s electing K data points from the dataset, either randomly or by choosing the first K points.
- ✓ Calculate the Euclidean distance between each data point in the dataset and the K centroids to determine which centroid is the closest to each data point.
- ✓ Assign each data point to its nearest centroid based on the distances computed in the previous step.
- ✓ Recalculate the centroids for each cluster by computing the average of all data points assigned to that cluster.
- ✓ Repeat steps 2 to 4 for a fixed number of iterations or until the centroids no longer change.

Euclidean Distance between two points in space:

$$d(p,q) = \sqrt{(\mathrm{q}_1 - \mathrm{p}_1)^2 + (\mathrm{q}_2 - \mathrm{p}_2)^2}$$

Assigning each point to the nearest cluster:

If each cluster centroid is denoted by ci, then each data point x is assigned to a cluster based on

$$\arg \min_{c_i \epsilon C} dist(c_i, x)^2$$

here dist () is the Euclidean distance

Finding the new centroid from the clustered group of points:

$$c_i = \frac{1}{|S_i|} \sum_{x_i \, \epsilon \, S_i} x_i$$

S_i is the set of all points assigned to the *ith* cluster.

4.2 PRINCIPAL COMPONENT ANALYSIS

Principal Component Analysis (PCA) is a dimensionality reduction technique commonly employed to condense large datasets by converting numerous variables into a smaller subset that retains the bulk of the original data's information. While this reduction inherently involves a sacrifice in precision, the key principle in dimensionality reduction is to exchange a degree of accuracy for simplicity. Smaller datasets offer advantages in terms of ease of exploration and visualization, simplifying data analysis and accelerating machine learning algorithms by eliminating extraneous variables. In essence, PCA's core concept is to minimize the number of data set variables while retaining maximal information.

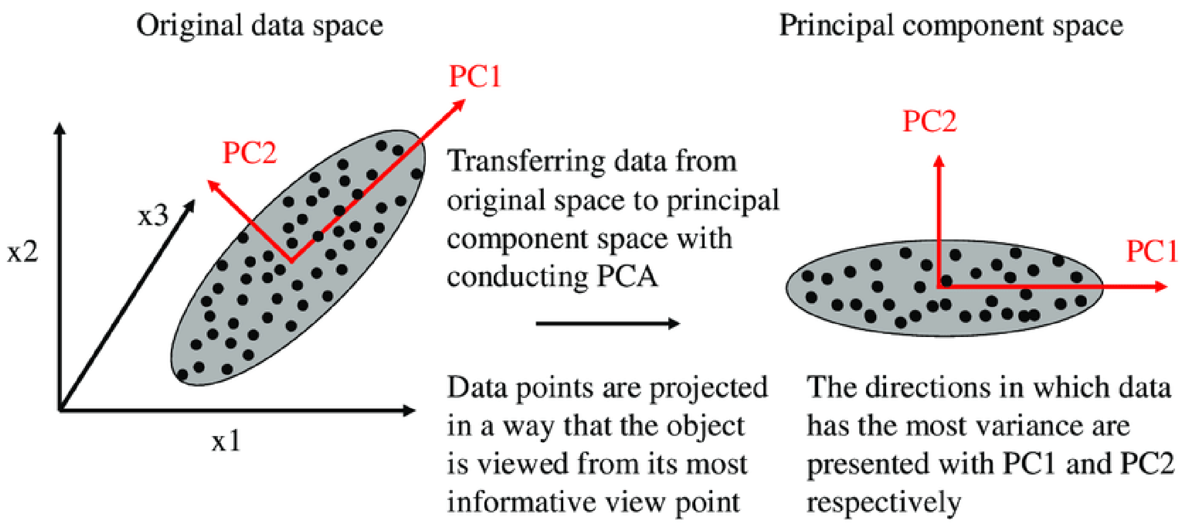


Figure 03: Data Space observation of Principal Component Analysis

(Source: analyticsvidhya.com)

In the new feature space of PCA, the goal is to identify distinctive properties that exhibit significant differences between classes. Unlike properties with low variance that render objects indistinguishable, PCA seeks characteristics that display the greatest divergence between classes, ultimately establishing the Principal Component Space. This process relies on mathematical concepts like Variance Matrix, Covariance Matrix, Eigenvectors, and Eigenvalues. Through these principles, PCA computes a set of eigenvectors and their corresponding eigenvalues, enabling the transformation of high-dimensional data into a reduced, more informative space. Detailed explanations of the Covariance Matrix, Eigenvalues, and Eigenvectors are provided in the subsequent section on the mathematical underpinnings of PCA.

To grasp the mathematical foundation of PCA, it's essential to comprehend the significance of Eigenvalues and Eigenvectors. Eigenvalues represent the new axes in the Principal Component space, while Eigenvalues quantify the variance each Eigenvector holds. Therefore, in dimensionality reduction, the choice of Eigenvectors to retain is based on their variance. Covariance and Variance, both denoting the data spread around its mean, play a vital role in PCA. Covariance examines relationships between two dimensions, such as hours studied and marks obtained. The formula for covariance is as follows:

$$\text{Cov}(X, Y) = \Sigma [(X_i - \bar{X}) * (Y_i - \bar{Y})] / (n - 1)$$

where X and Y represent the two dimensions, X_i and Y_i denote individual data points, $\bar{X}$ and $\bar{Y}$ are the means of the respective dimensions, and 'n' is the number of data points. This mathematical foundation is fundamental for PCA's dimensionality reduction process.

PCA involves several steps:

- ✓ Standardize the continuous variables within the dataset.
- ✓ Calculate the covariance matrix to reveal relationships between variables.
- ✓ Compute the eigenvalues and eigenvectors of the covariance matrix to identify principal components.
- ✓ Decide which principal components to retain for analysis based on their variance, typically using a Scree Plot.
- ✓ Transform the data to align with the axes of the selected principal components.

4.3 RANDOM FOREST

Random forest is a supervised learning algorithm that creates an ensemble of decision trees using the bagging method. It combines multiple decision trees to improve prediction accuracy and stability. In essence, it leverages the power of multiple models for better results. Random Forest is a supervised learning algorithm that employs ensemble learning, a technique involving multiple algorithms or repeated use of a single algorithm to enhance model performance. In this approach, it constructs numerous decision trees using randomly selected data points and then averages their predictions. By aggregating the results of these decision trees, it ultimately determines the final class for a given test object. This ensemble tree-based learning algorithm aims to improve prediction accuracy and robustness.

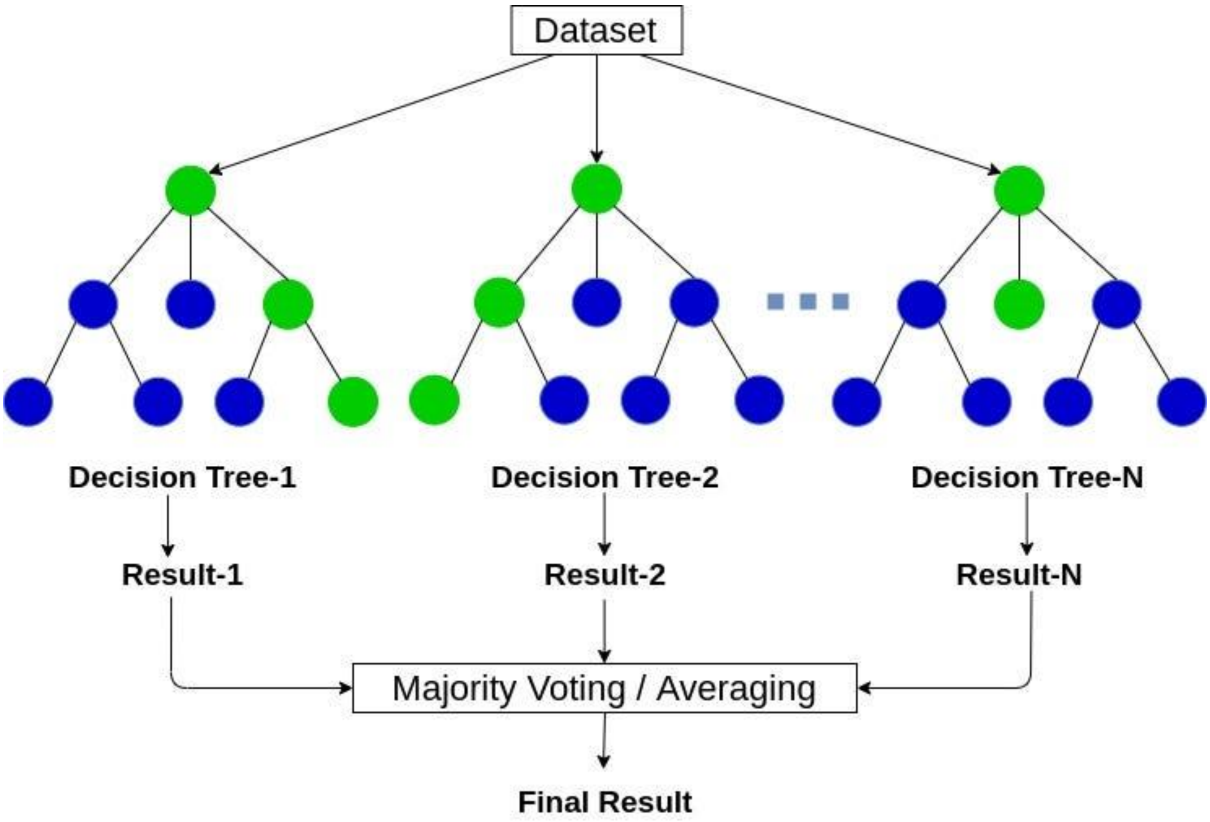


Figure 04: Structure of Random Forest

(Source: medium.com)

The Random Forest algorithm operates through the following steps:

- ✓ It begins by randomly selecting samples from a given dataset.
- ✓ For each of these samples, the algorithm constructs an individual decision tree and obtains a prediction result from each tree.
- ✓ Voting is then carried out for all the predicted results.
- ✓ Finally, the prediction result that receives the most votes is chosen as the ultimate prediction.

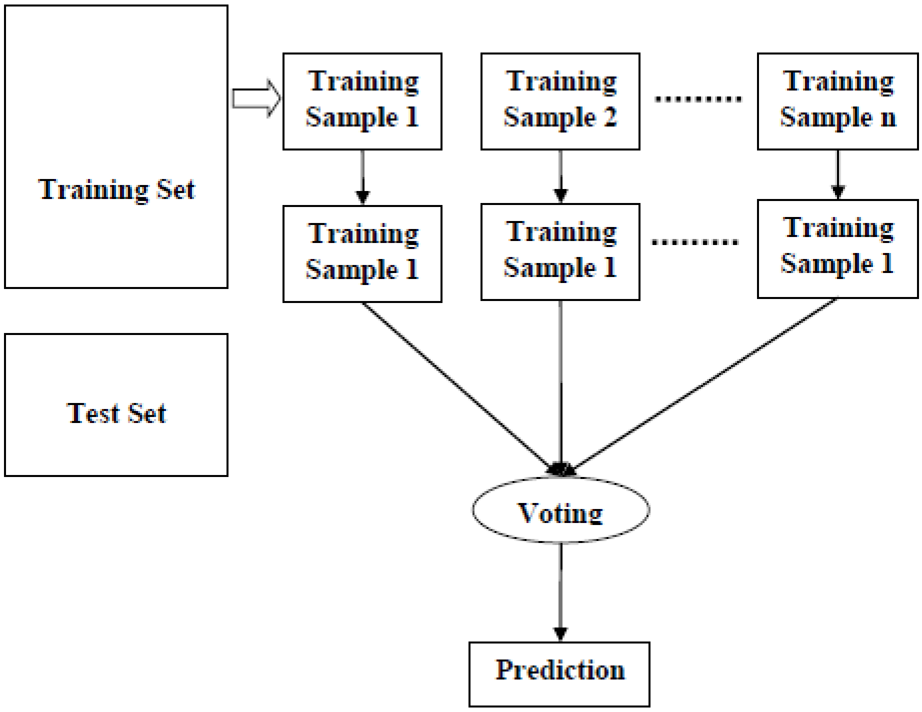


Figure 05: Working of Random Forest Classification algorithm

(Source: medium.com)

5 CHAPTER 5: DATASET AND FRAMEWORKS

This section delves into the technologies and frameworks employed in conducting experiments for the thesis, outlining the entire life cycle of the anomaly detection process in cryptocurrency transactions. In addition to this, it provides visual representations for the exploratory analysis of the data while detailing the creation and preprocessing of the dataset to facilitate algorithm modelling.

5.1 TECHNOLOGY AND FRAMEWORKS

Python is the programming language which is used in this thesis to conduct the data preparation and analysis. It is also the foundation to construct machine learning algorithm for anomaly detection in cryptocurrency transaction. Python is a high-level, versatile, and interpreted programming language known for its readability and easy to use. Numerous libraries are used to formulate the experiment. The major ones are as follows:

NumPy: It serves as the core library for array computation in Python, essential for a wide range of computational tasks.

Pandas: It provides strong data structures to facilitate data analysis, and it is primarily utilized for the management and manipulation of data.

Scikit-Learn: It incorporates a range of standard machine learning algorithms and tools for model evaluation, including K-means, ROC Curve, F1-Score, and others.

Matplotlib: It is a data visualization tool for displaying data.

Seaborn: This tool is employed for visualizing statistical data.

5.2 DATA SOURCE AND DATASET

The dataset for this thesis is created using raw data from the Bitcoin blockchain, which was obtained by syncing with the Bitcoin client software over the internet. The Bitcoin blockchain stores all transaction data in a public ledger, and the currency unit used is Bitcoin (BTC). This ledger contains records of all Bitcoin transactions from the inception of the network to the present. According to Blockchain.com (2019), there are approximately 450 million transactions recorded in the Bitcoin ledger. Each transaction can involve multiple sender and receiver addresses, and it's worth noting that a single user can possess multiple addresses. Importantly, all users remain anonymous as there is no personal information linked to any of the addresses.

The experiment is initiated with four datasets. Each consists of its own variables and relations of Bitcoin transactions. First dataset is named as "blockhash" which contains the information of Bitcoin transaction block. In total, it contains 2703 transaction blocks. The variables are indicated as follows:

- ✓ blockID: ID of block used in our dataset
- ✓ bhash: blockhash (block identifier in the blockchain)
- ✓ btime: creation time of the block
- ✓ txs: number of transactions

	blockID	bhash	btime	txs
0	327872	000000000000000001BECC903BF44978DEDE7C0429B08AF...	1414771239	288
1	327873	00000000000000000008F1C9DC87C18D50D6AAC82934DF4...	1414771715	256
2	327874	000000000000000001A80CF302D79ED1322AFB14F9E9455...	1414771566	176
3	327875	00000000000000000061E549AB51D10C89599C8BE690E6E...	1414771623	121
4	327876	000000000000000001068FE142FA57C1505BAE0DD2FC148...	1414772032	490

Figure 06: Dataset "blockhash"

The second one is named as "tx" which represents the information of a transaction. In total, it contains 1,048,575 transactions. The variables are indicated as follows:

- ✓ txID: transaction ID
- ✓ blockID: ID of block used in our dataset
- ✓ n_inputs: number of transaction input
- ✓ n_outputs: number of transaction output
- ✓ btime: creation time of the block

	txID	blockID	n_inputs	n_outputs	btime
0	50230631	327872	18	2	1414771239
1	50230632	327872	1	2	1414771239
2	50230633	327872	27	1	1414771239
3	50230634	327872	1	2	1414771239
4	50230635	327872	331	1	1414771239

Figure 07: Dataset "tx"

The third dataset is named as "txin" which represents the input of a transaction. In total, it contains 1,048,575 transactions. The variables are indicated as follows:

- ✓ txID: transaction ID
- ✓ addrID: ID of sending address
- ✓ value: integer sum in Satoshis ($1e^{-8}$BTC)

	txID	send_addrID	value
0	51575940	422	590000
1	51575940	422	3970000
2	51576100	422	13640000
3	51576102	422	12520000
4	51576103	422	11400000

Figure 08: Dataset "txin"

The fourth dataset is named as "txout" which represents the output of a transaction. In total, it contains 1,048,575 transactions. The variables are indicated as follows:

- ✓ txID: transaction ID
- ✓ addrID: ID of receiving address
- ✓ value: integer sum in Satoshis ($1e^{-8}$BTC)

	txID	receive_addrID	value
0	50511380	112	5480
1	50511375	139	5480
2	50283564	161	5480
3	50511381	165	5480
4	50511376	216	5480

Figure 09: Dataset "txout"

5.3 DATA PREPARATION

Data Preprocessing is the integral part that needs to be executed before applying the dataset to the machine learning algorithms. Preprocessing the data, such as cleaning and transforming, ensures it does not mislead the analysis. The first approach toward data cleaning is to remove any rows that contain missing data points. There are no missing values in the Bitcoin transaction datasets.

The main objective of this experiment is to identify anomaly among the Bitcoin transaction blocks. Therefore, the data preprocessing steps are needed to modify in such a way that various characteristics of the transaction blocks can be extracted. The data preprocessing steps are designed as follows:

Step 1: The dataset "tx", "txin" and "txout" are merged based on their common feature "txID"**.** The datasets are combined to accumulate the transaction features in a single dataset.

	txID	blockID	n_inputs	n_outputs	btime	send_addrID	value_x	receive_addrID	value_y
0	50230639	327872	3	2	1414771239	6078391	1262832	15157877	1001311
1	50230641	327872	9	19	1414771239	12389926	30000000	27466155	1000240
2	50230641	327872	9	19	1414771239	19092393	1000913	27466155	1000240
3	50230641	327872	9	19	1414771239	26300432	149900000	27466155	1000240
4	50230676	327872	1	2	1414771239	3886651	150000	3886651	96340

Figure 10: Accumulation of Datasets

Step 2: As the main objective is to assimilate the data based on transaction blocks, the pivot table is used to derive four characteristics of individual blocks.

	n_inputs	n_outputs	value_x	value_y
blockID				
327872	57010	1453	17922561000	5326933667
327873	16886	430	9626161187	199267946940
327874	4498	4621	6257128442	7418868564
327875	7585	829	12482161809	73029654331
327876	12104	7304	245349326819	424617793178

Figure 11: Processed Dataset

Step 3: Finally, the pivot table is merged with dataset "blockhash" based on "blockID" column and dropped unnecessary information.

	blockID	n_inputs	n_outputs	send_value	receive_value	btime	txs
0	327872	57010	1453	17922561000	5326933667	1414771239	288
1	327873	16886	430	9626161187	199267946940	1414771715	256
2	327874	4498	4621	6257128442	7418868564	1414771566	176
3	327875	7585	829	12482161809	73029654331	1414771623	121
4	327876	12104	7304	245349326819	424617793178	1414772032	490

Figure 12: Final Dataset

5.4 EXPLORATORY DATA ANALYSIS

The exploratory data analysis involves employing a range of visualizations to gain deeper insights into the data and comprehend the relationships between its different aspects. This analysis was beneficial not only in identifying any imbalance between data classes but also in examining the correlations among the data features. Moreover, it contributes to our understanding of the types of data transformations that could be applied to prepare it for modelling purposes.

After the data preparation, we finalize the Bitcoin transaction block dataset with six features which are named as "n_inputs", "n_outputs", "send_value", "receive_value", "btime" and "txs". The final dataset has 1867 rows and 7 columns including the distinguishing "blockID".

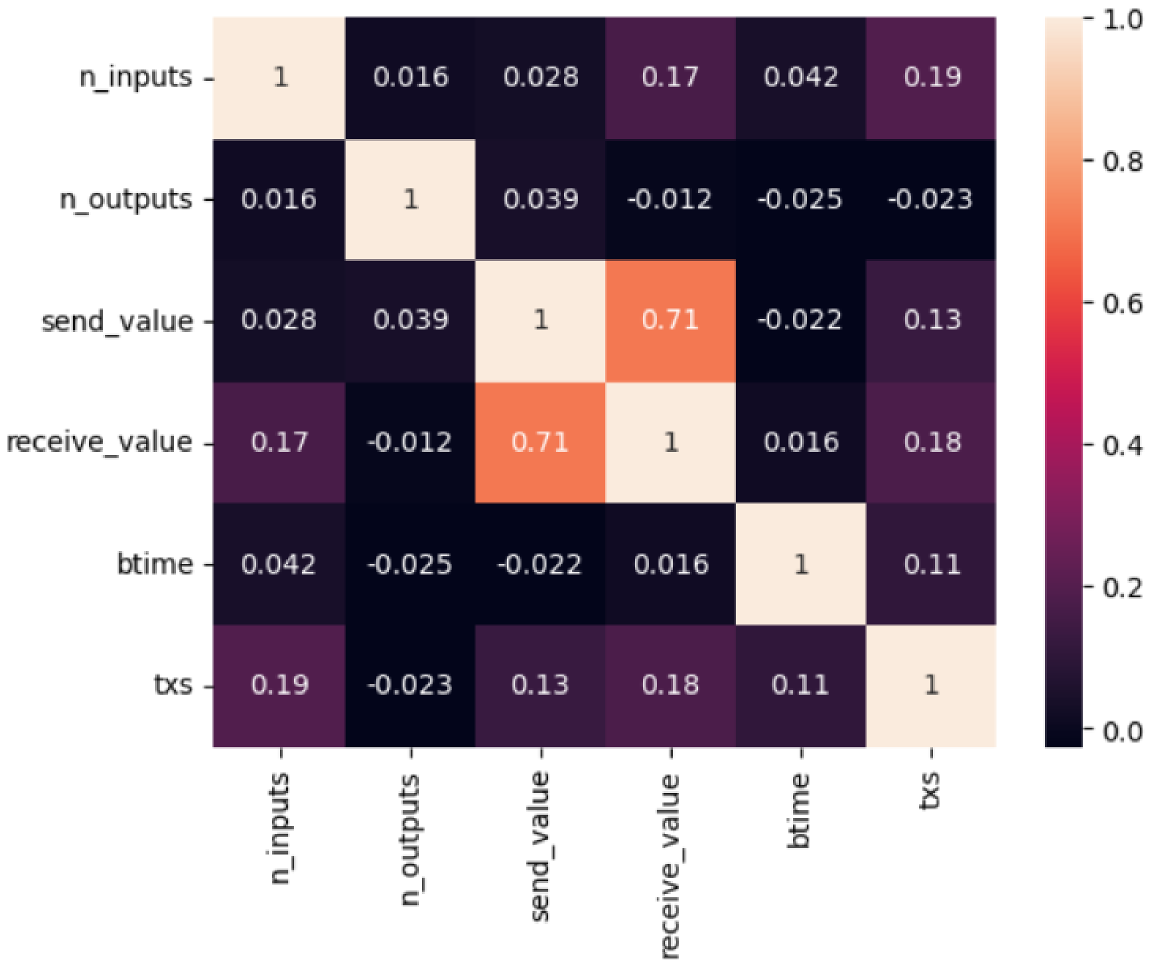


Figure 13: Correlations of different variables

Firstly, a correlation heatmap is used to analysis the relationship among the variables. It depicts each variable both on the rows and columns of the heatmap, forming a symmetric matrix. The cells within the heatmap display color-coded values that signify the strength and direction of correlations between variables. It highlights a strong correlation of 0.71 between send_value and receive_value which denotes the tendency that if an individual block has high value of sending Bitcoin, it also incorporates the possibility of high amount of receiving Bitcoins as well. However, other characteristics shows little or no correlations among themselves. Similar trends are also showing through the pair plots but more visible way. Relational diagrams are also showing that majority of the transactions are grouped together among themselves and only few of them are depicted as an outlier. It is the main motivation to use unsupervised K-means clustering to cluster the similar transaction blocks for labelling the dataset.

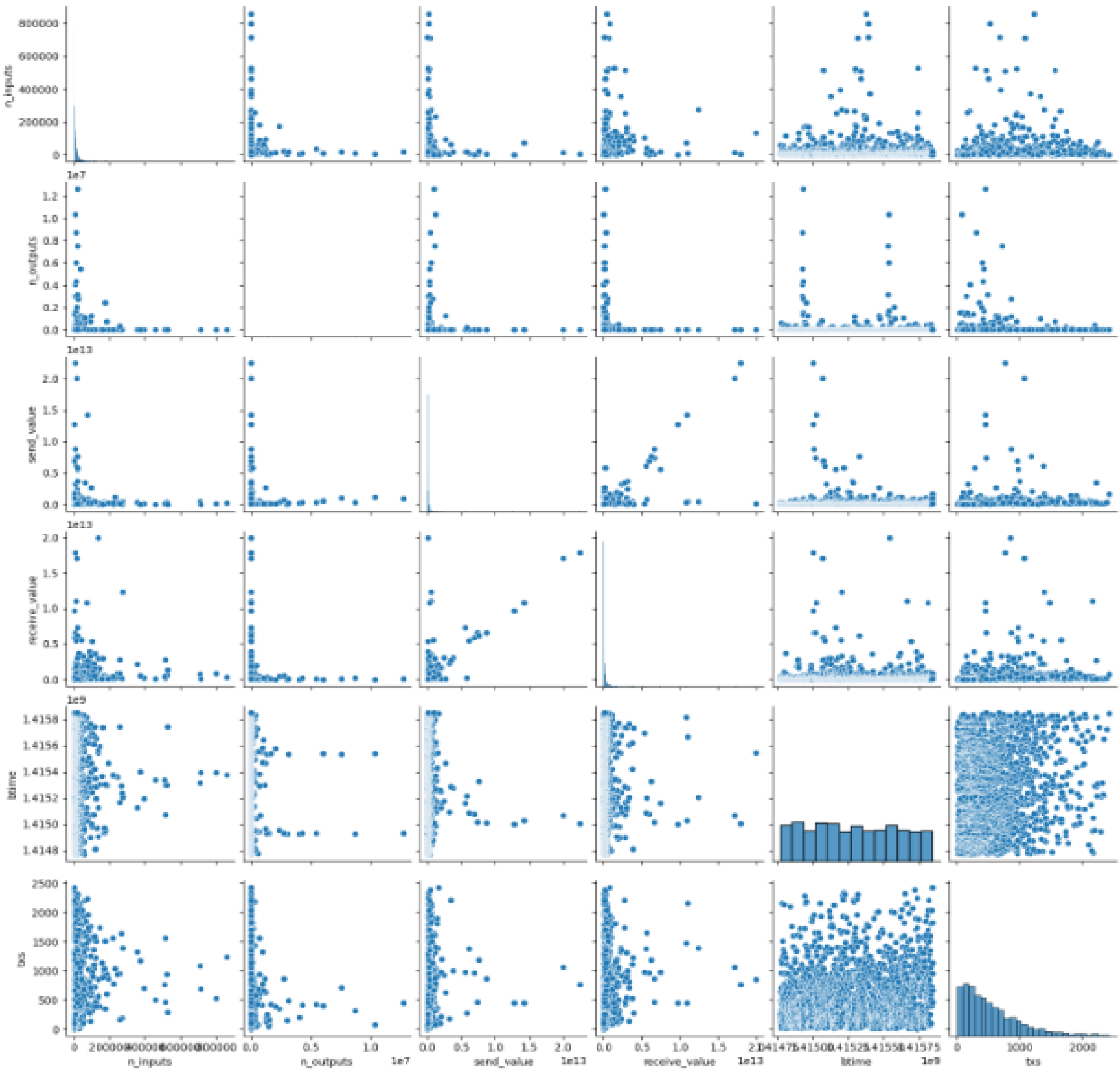


Figure 14: Heatmap diagram of the correlations of variables

Finally, we focus on the distribution of each variable. In this case, the distribution plot and box plot are used to diagnose the characteristic of individual variable. A distribution plot is a data visualization technique commonly used in data analysis and statistics. The primary purpose of a distribution plot is to visualize the distribution of a dataset, allowing to explore the underlying structure of the data and gain insights into its central tendencies, variability, and the presence of anomalies or outliers. Simultaneously, Box plot also provides a summary of the data distribution within each category, showing the median, quartiles, and any potential outliers.

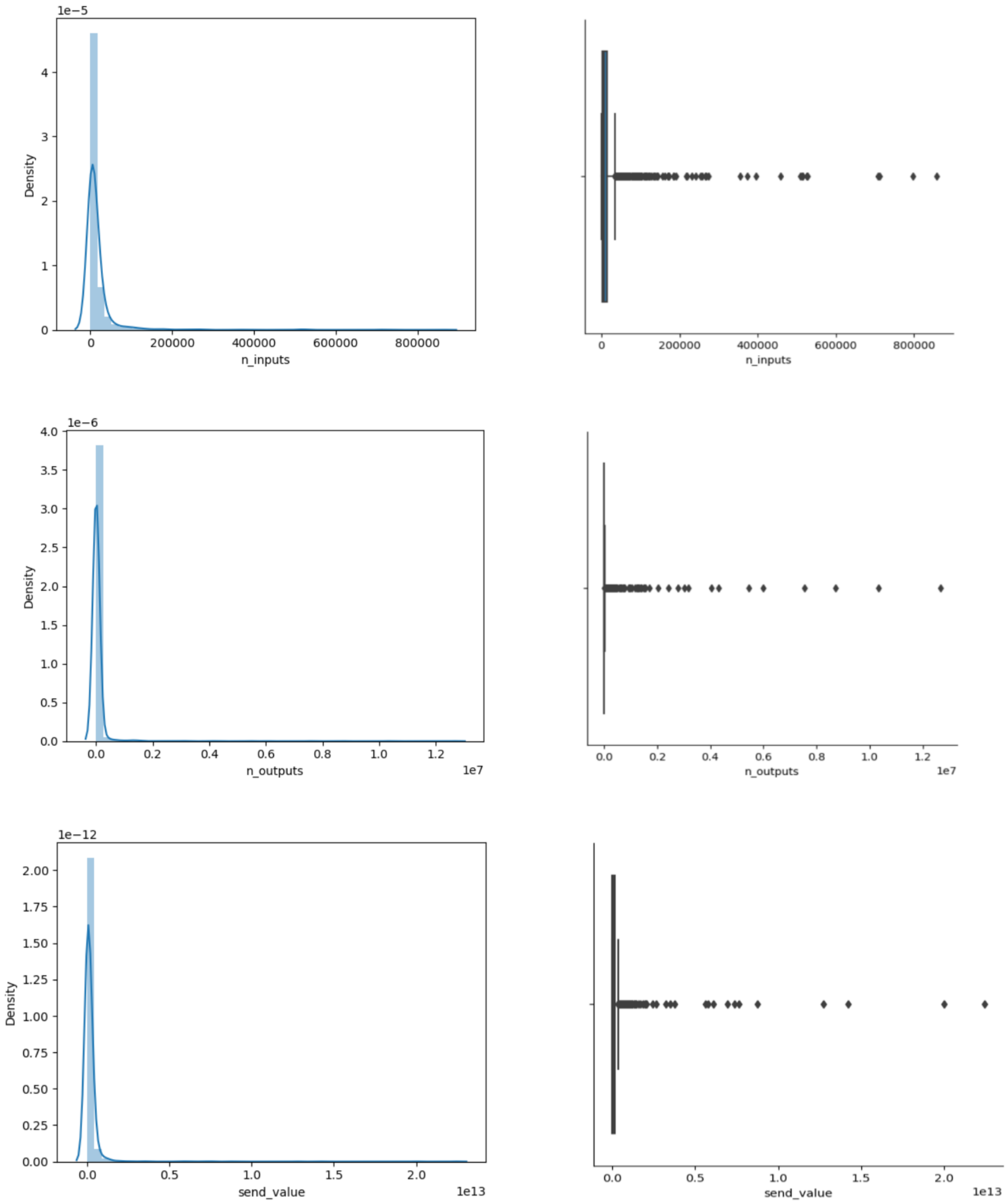


Figure 15: Data Distribution of n_inputs, n_outputs and send_value

Both the distribution plot and box plot of variables n_input, n_output, send_value and receive_value indicate that there is a similar characteristic among most of the transaction blocks which justifies the unsupervised K-means clustering method of this thesis. Distribution plots of these variables are highly right-skewed or positively skewed. Similar behaviours are also depicted in the box plot of these variables. In addition, btime which denotes the creation time of the transaction block are equally distributed as the formation methodology of each block are almost similar. Further, transaction (txs) variable is also right-skewed but the skewness is less extreme compared with n_input, n_output, send_value and receive_value.

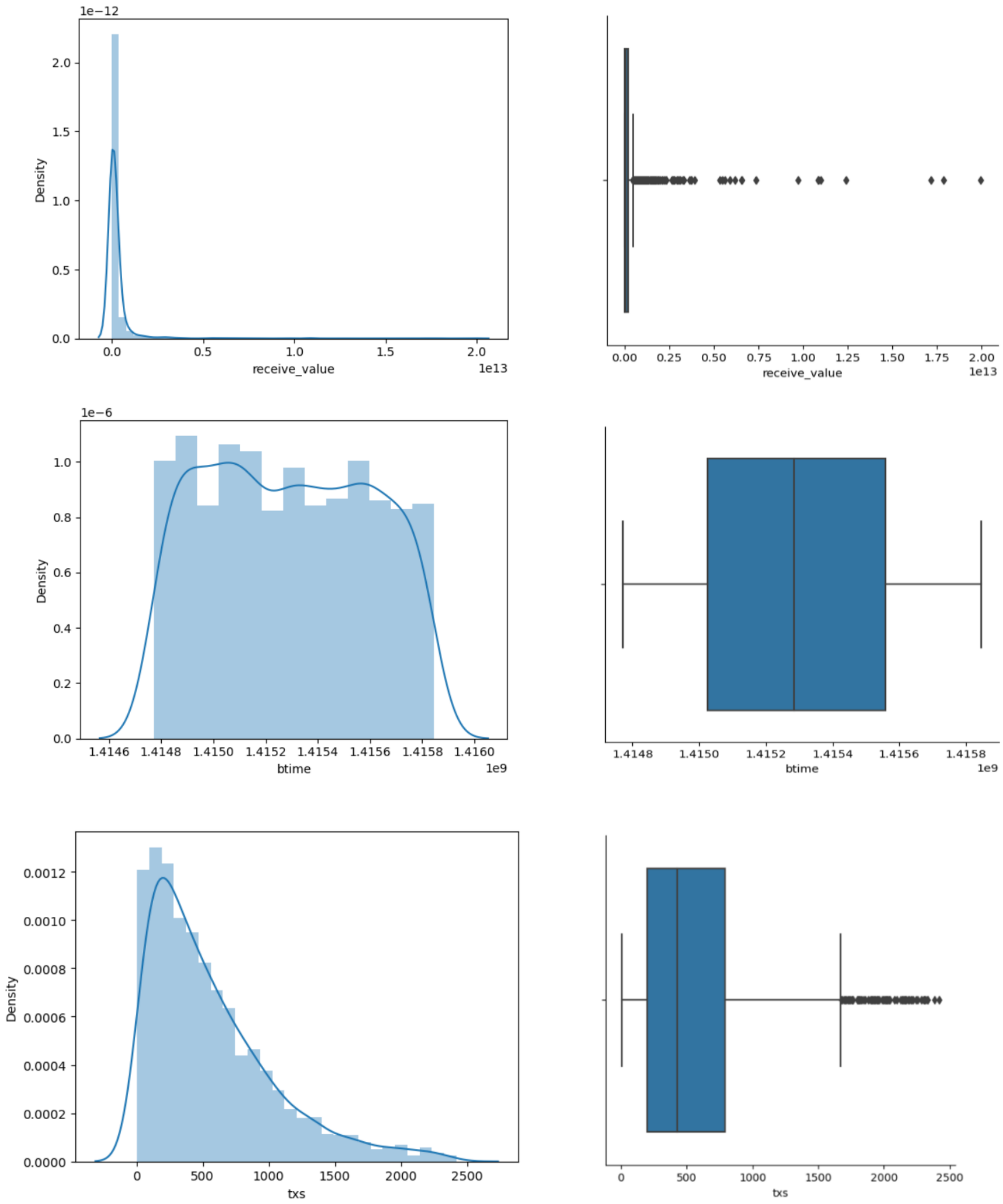


Figure 16: Data Distribution of receive_value, btime and txs

6 CHAPTER 6: METHODOLOGY

The ultimate objective of this methodology is to identify anomalous transaction block. Because of the limitation of labelled data, the experiment method initiates with the combination of the unsupervised ML and the supervised ML algorithm. K-means clustering is implemented as unsupervised ML model to identify normal and anomalous block. With the assistance of this identification, Principal Component Analysis and Random Forest as supervised ML algorithms build the anomaly detection model in this experiment.

6.1 K-MEANS CLUSTERING ALGORITHM

K-means clustering is a popular unsupervised machine learning algorithm used for partitioning a set of data points into distinct, non-overlapping groups or clusters. The algorithm aims to group similar data points together and separate dissimilar ones based on their features. The very first task in this algorithm is to find the optimal number of clusters. The selection of the optimal number of clusters in K-means clustering is a crucial decision and should not be arbitrary. It involves calculating the mean distances of each data point within a cluster from its centroid. The Within-Cluster-Sum-of-Squares (WCSS) method is employed for this purpose, which quantifies the sum of the squared distances between data points and their respective cluster centroids. The objective is to minimize these distances, typically measured using the Euclidean distance, by iteratively adjusting the number of clusters until a minimum sum of distances is achieved.

6.1.1 ELBOW METHOD

To determine the ideal number of clusters using the elbow method, these steps are followed:

Step 1: Apply K-means clustering to the dataset, varying the number of clusters (K) from 1 to 10.

Step 2: Calculate the Within-Cluster-Sum-of-Squares (WCSS) for each K value.

Step 3: Create a graphical plot, showing the relationship between WCSS values and the corresponding number of clusters (K).

Step 4: Identify the point on the plot where there is a noticeable bend, resembling an elbow joint; this point is considered the optimal K value.

From the Elbow method, the optimal number of clusters can be identified as 2 which paves the way to indicate the clusters as normal and anomalous transaction blocks through unsupervised K-means clustering algorithm.

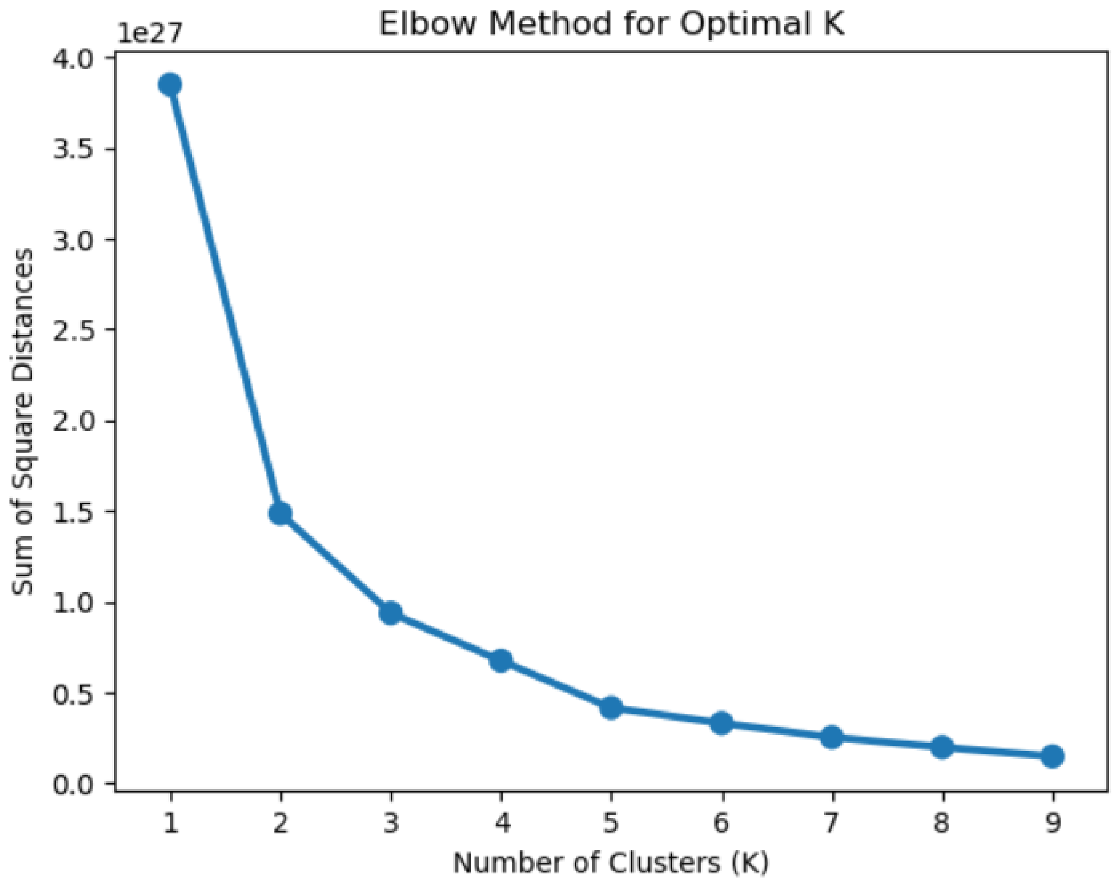


Figure 17: Determination of the optimal number of clustering by Elbow Method

Furthermore, we have applied K-means clustering considered all 6 variables. 1853 transaction blocks have identified as normal labelled data and 14 transaction blocks as anomaly dataset. The process for K-means clustering is relatively simple:

Step 1: Determine the desired number of clusters, denoted as 'k'.

Step 2: Initially, assign a centroid randomly to each of the 'k' clusters.

Step 3: Calculate the distances between all data points and each of the 'k' centroids.

Step 4: Assign each data point to the nearest centroid.

Step 5: Update the centroid positions by calculating the mean of all data points within each cluster.

Step 6: Iterate through steps 3 to 5 until the centroids no longer change their positions.

6.1.2 SILHOUETTE SCORE

The silhouette coefficient, specifically when applied to K-means clustering, serves as an indicator of how much a data point resembles the cluster it belongs to (cohesion) relative to other clusters (separation). For a particular data point, the calculation of the silhouette coefficient is as follows:

$$S(i) = \frac{\mathrm{b(i)} - a(i)}{\max\ \{a(i),\ b(i)\}}$$

- ✓ S(i) denotes the silhouette coefficient of the data point i
- ✓ a(i) is the average between i and all other data points in the cluster to which i belongs
- ✓ b(i) is the average distance from i to all clusters to which i does not belong

The average_silhouette for every k can be calculated as follows:

$$\text{AverageSilhouette} = \text{mean}\{S(i)\}$$

In Python, calculating the Silhouette score is straightforward and can be done using the "metrics" module within the scikit-learn (sklearn) library.

Step 1: Select a range of values of k

Step 2: Plot Silhouette coefficient S(i) for each value of K

Step 3: Plot the graph between average_silhouette and K

While calculating the Silhouette Coefficient, the value of the silhouette coefficient is between [-1,1]. The Silhouette Coefficient is 0.97 which illustrates that the data point i is very compact within the cluster to which it belongs and far away from the other clusters. Further, the Silhouette Score is maximized at K = 2, so the optimal cluster of 2 is justified from the diagram.

6.2 SUPERVISED PRINCIPAL COMPONENT ANALYSIS ALGORITHM

Principal Component Analysis (PCA) is a commonly used machine learning method aimed at reducing the dimensionality of extensive datasets. Its primary goals are to enhance interpretability and, simultaneously, minimize the loss of information. Usually, visualizing data with high dimensionality can be challenging. To address this, we can employ PCA to identify the initial two principal components and create a single scatter plot in this newly established two-dimensional space. However, prior to this, we must standardize our data to ensure that each feature has a uniform variance of one.

PCA in Scikit-Learn follows a process akin to other preprocessing functions within the Scikit-Learn library. We create a PCA object, determine the principal components with the fit method, and subsequently apply rotation and dimensionality reduction. Additionally, we have the option to specify the number of components to retain when initializing the PCA object. Using just these two components, it becomes evident that we can effectively distinguish between the two classes. In this distinguishing method, we have used the labelled data from K-means clustering. From the visualization, it is evident that the label from K-means clustering can be justified by Principal Component Analysis. Therefore, we can justify the anomalous Bitcoin transaction blocks through this anomaly detection model.

Regrettably, while dimensionality reduction is a powerful tool, it comes with the drawback of making it less straightforward to comprehend the meaning of these components. These components are essentially combinations of the original features and are accessible as an attribute of the PCA object once it's fitted.

The heatmap below, along with the colour bar, illustrates the correlations between the principal component and the various features. A heatmap is a graphical representation of data where values in a matrix are depicted using a range of colours. In this visualization, each cell's colour intensity is proportional to the value it represents. From the figure 18, it is highlighted that n_output, send_value and receive_value are more correlated with the principal components. Though, n_input and txs are less correlated, btime which denotes as block creation time has the least contribution on the principal components.

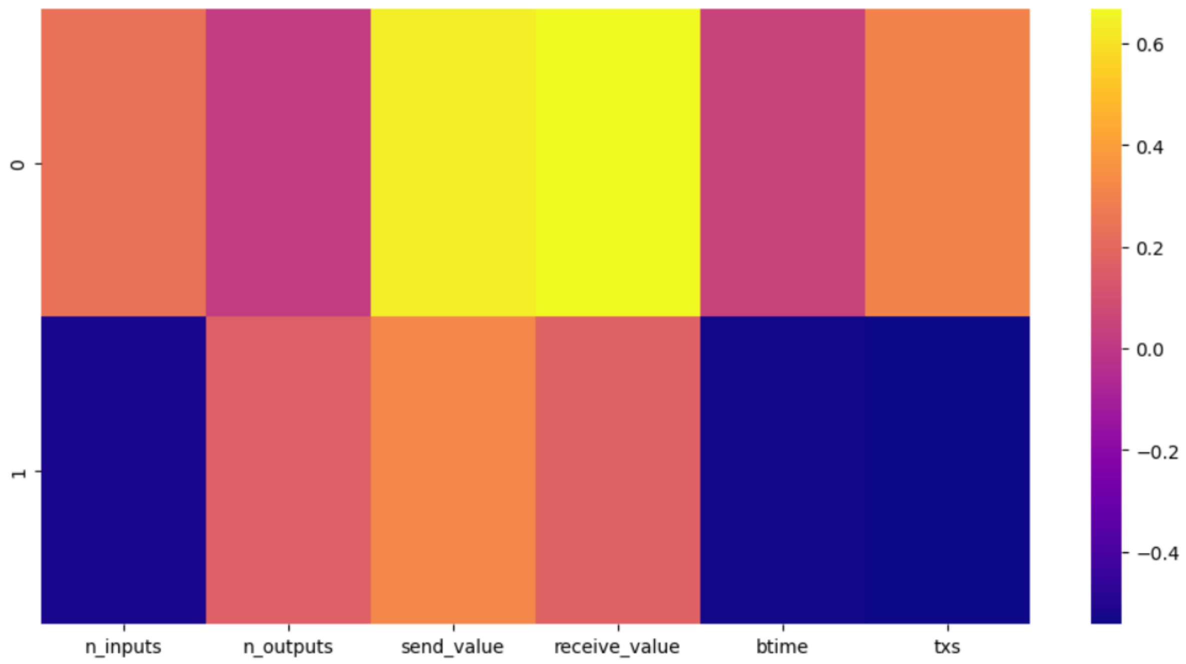


Figure 18: Variable evaluation by Principal Component Analysis

6.3 RANDOM FOREST ALGORITHM

Random Forest is a powerful ensemble learning algorithm used in supervised machine learning. It's a versatile method for both classification and regression tasks. Random Forest builds a "forest" of decision trees, where each tree independently makes predictions. The final prediction is determined by taking a majority vote for classification. Random Forest is known for its robustness and ability to handle large datasets with numerous features. It helps reducing overfitting and increasing model accuracy.

Step 1: Importing random forest classifier library from scikit-learn and setting the random seed

Step 2: Splitting the data within training and test set. We use 75% data as training set and remaining 25% as test dataset. Therefore, we have 1401 data as training set and other 466 as test dataset.

Step 3: Creating the random forest classifier and fitting the training set and target values.

Step 4: Applying the trained classifier to the test set which returns the prediction.

Step 5: Viewing the predicted probabilities of the observation using predict_proba function.

Step 6: Replacing binary digit with “Normal” and “Suspicious” for viewing the result of the observation.

6.3.1 CONFUSION MATRIX

The confusion matrix is a widely utilized technique for assessing the performance of classification models, particularly in the context of machine learning classification algorithms like logistic regression. This matrix, typically a 2x2 configuration, is employed in binary classification scenarios to gauge the model's accuracy by comparing its predictions to actual data points. In practice, logistic regression models are trained using in-sample data, and their accuracy is subsequently assessed with out-of-sample data that the model hasn't encountered before. The confusion matrix facilitates the comparison of actual out-of-sample data with the model's predictions, presenting the results in a four-class (2x2) matrix format. It is a fundamental and extensively available tool for evaluating the performance of machine learning classification algorithms in the field.

The confusion matrix comprises four segments: True Positive, True Negative, False Positive, and False Negative. These elements are defined and explained as follows. The explanation and formula for the confusion matrix are sourced from the provided reference.

True Positive: True Positive (TP) occurs when the model correctly predicts out-of-sample observations as belonging to the positive class, where both the predicted and actual values are 1, signifying accurate predictions of the out-of-sample positive class.

Predicted Class

Actual Class	Predicted Class	Predicted Class
Negative (0)	True Negative	False Positive
Positive (1)	False Negative	True Positive

Figure 19: Confusion Matrix Table

True Negative: True Negative (TN) is when the model accurately predicts out-of-sample observations as part of the negative class. In this scenario, both the predicted and actual values are 0, indicating precise predictions of the out-of-sample negative class.

False Positive: False Positive (FP) occurs when the model incorrectly predicts observation values as part of the positive class, but the actual value is negative. In this situation, the model's predictions falsely categorize the negative class as the positive class, often referred to as a Type I error. The predicted label (class) is 1, while the actual label (class) is 0.

False Negative: False Negative (FN) occurs when the model incorrectly predicts observation values as part of the negative class, but the actual value is positive. In this scenario, the model's predictions falsely categorize the positive class as the negative class, often referred to as a Type II error. The predicted label is 0, while the actual label is 1.

Following the computation of the confusion matrix by machine learning classification algorithms, such as logistic regression, a quantitative evaluation report can be generated. This report, known as the classification report, is produced using the methodology outlined below.

Accuracy: Accuracy refers to the model's overall effectiveness, expressed as a percentage. It indicates the model's ability to make accurate predictions by revealing the proportion of correct predictions out of all the predictions made by the model. It calculates as:

$$\text{Accuracy} = \frac{\text{True Positive} + \text{True Negative}}{\text{True Positive} + \text{True Negative} + \text{False Positive} + \text{False Negative}}$$

Precision: In a binary class Logistic regression model, there are two classes: positive and negative. Here, the positive class represents a bull market, and the negative class represents a bear market. When focusing on the positive class, precision measures the accuracy of the model's predictions for the bull market. Precision is determined by dividing the total number of correctly predicted positive class values by the overall number of positive class predictions made by the model. It computes as follows:

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

Recall: The recall or sensitivity score assesses the proportion of positive class predictions made by the model out of all the positive observations present in the dataset. In other words, it measures how effectively the model identifies the bull market by considering all the actual bull market observations in the out-of-sample data. Typically, recall is employed to evaluate the performance of the positive class, which is why it is also referred to as the sensitivity score. It can calculate by the following:

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

F1-Score: The F1 score is a widely used metric in machine learning classification that merges a model's precision and recall into a single measure. It calculates the average of both precision and recall, assigning equal importance to each. The F1 score offers a balanced evaluation of classifier performance by taking both precision and recall into account. A high F1 score signifies strong classifier performance with minimal occurrences of false positives and false negatives. Mathematically, the F1 score is calculated as follows:

$$\text{F1-Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} * \text{Recall}}$$

The F1 score falls within the range of 0 to 1, with a score of 1 indicating perfect precision and recall, while a score of 0 signifies the model's poor performance.

7 CHAPTER 7: THESIS DISCUSSION

This paper fabricates unsupervised ML algorithm K-means clustering and supervised Principal Component Analysis and Random Forest to differentiate anomalous transaction blocks. The evaluation and discussion of the dataset and each ML model is described in this chapter.

7.1 DATASET EVALUATION

During the exploratory analysis of the dataset, we initially delved into the dataset's characteristics using various visualizations. We began by examining the relationships between variables through a correlation matrix. Notably, the variables "send_value" and "receive_value" displayed a strong correlation, while the majority of other variables exhibited slight correlations. To gain a more graphical perspective on these correlations, we employed pair plots. These plots allowed us to visualize the relationships between variables. Furthermore, we used distribution plots and box plots to illustrate the distribution patterns of each variable. The overarching observation from these plots was that most of the data points clustered closely together, with only a few scattered outliers. This observation substantiates the central idea of the thesis, which involves utilizing unsupervised machine learning algorithms to categorize data based on their inherent clustering tendencies.

7.2 K-MEANS CLUSTERING EVALUATION

Evaluation with the Elbow Method

The main goal of employing K-means clustering in this study is to categorize the Bitcoin transaction block dataset into two distinct groups: "Normal" and "Suspicious." The primary aim is to establish a clear separation of these two categories within the dataset. This critical task involves determining the most appropriate number of clusters that effectively differentiate "Normal" from "Suspicious" transactions. To achieve this, the researchers have utilized the Elbow method, a fundamental technique in machine learning and statistics. The Elbow method plays a crucial role in identifying the ideal number of clusters when applying clustering algorithms, particularly K-means clustering, to a dataset. Following the application of the Elbow method, it has become evident that the most suitable number of clusters for this specific Bitcoin transaction block dataset is two. This finding serves as the basis for labelling the dataset as either "Normal" or "Suspicious," aligning with the two clusters that have been revealed. The K-means clustering algorithm further substantiates this result by highlighting the presence of 1853 blocks categorized as "Normal" and 14 blocks classified as "Anomalous" or "Suspicious." In essence, the core objective of this experiment revolves around the establishment of these two distinct clusters, enabling the accurate classification of the Bitcoin transaction block dataset into these essential "Normal" and "Suspicious" categories.

Evaluation with the Silhouette Score

The silhouette score is a measurement employed to gauge the effectiveness of clusters generated by clustering algorithms like k-means. It assesses how similar data points within the same cluster are to one another in comparison to the nearest neighbouring cluster. Essentially, it quantifies both the distinctness of clusters and the cohesion of data points within those clusters.

When evaluating a clustering using this metric, a silhouette score of 0.97 for K-means indicates that data points are excellently suited to their respective clusters and not as well-suited to neighbouring clusters. This high score signifies the presence of well-defined and distinct clusters, a hallmark of a successful clustering. Additionally, the optimal number of clusters can be determined by examining the silhouette scores across different cluster counts. The silhouette score offers insights into the suitability of various cluster numbers for the given dataset.

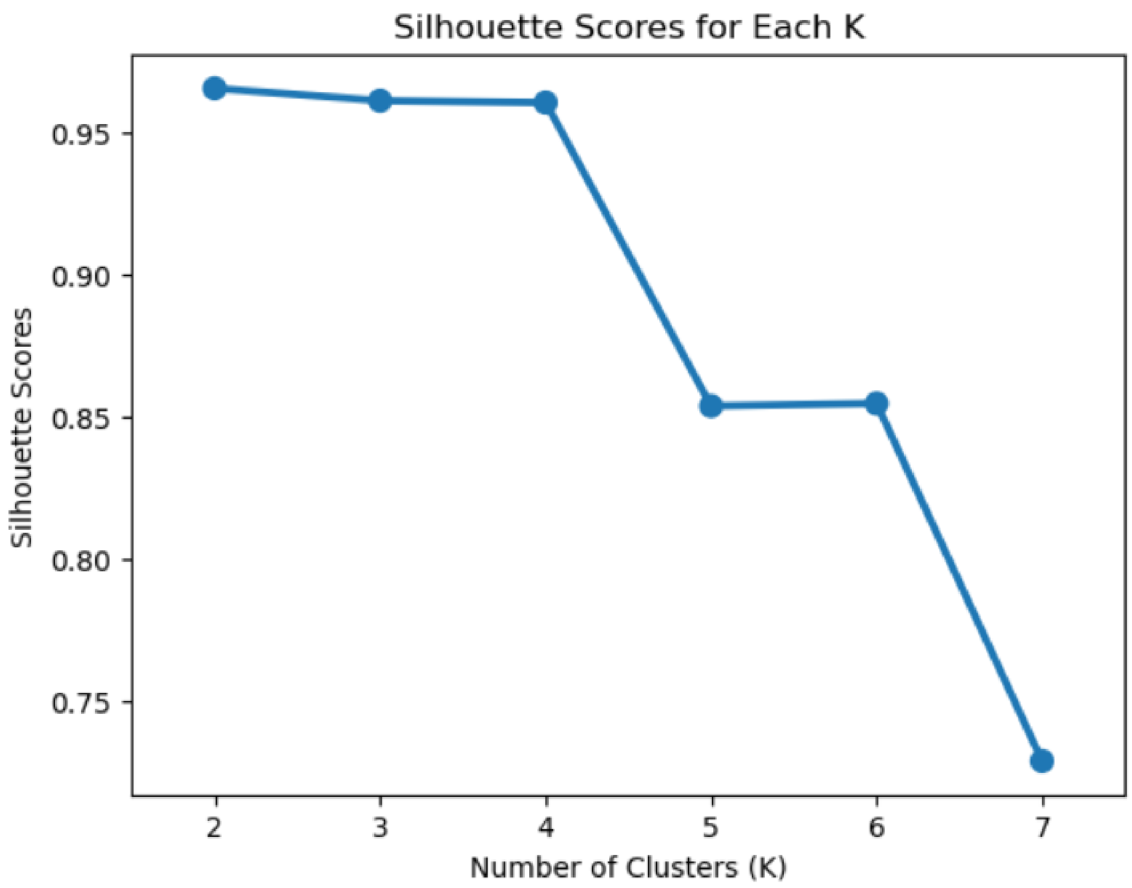


Figure 20: Silhouette Score for the number of clusters

7.3 SUPERVISED PRINCIPAL COMPONENT ANALYSIS EVALUATION

Principal Component Analysis (PCA) is a method employed in both statistics and machine learning to condense high-dimensional data into a lower-dimensional format while preserving the key data patterns and relationships. Its main purpose is to simplify intricate data, enabling easier analysis, visualization, and modelling.

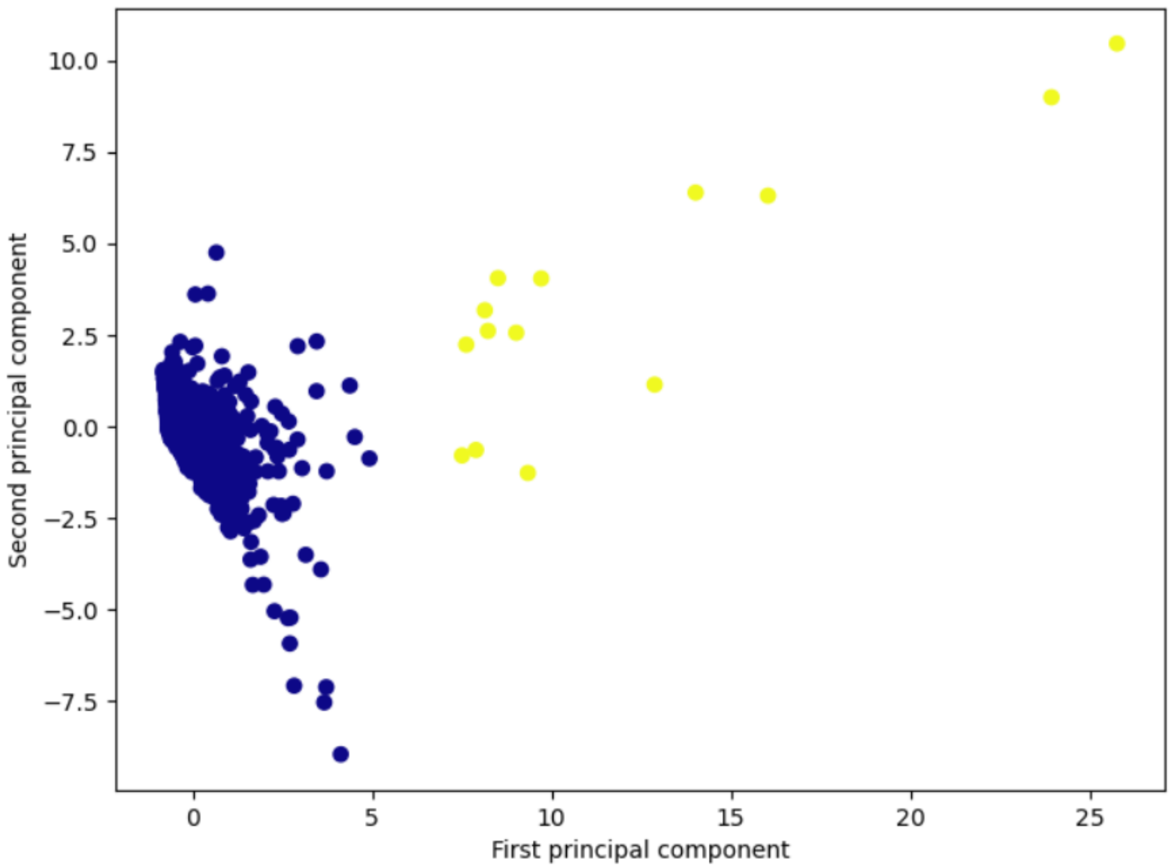


Figure 21: Visualization of Principal Component Analysis

In this particular experiment, the dataset's dimensionality is reduced from 6 variables to just 2 by employing the PCA algorithm. Subsequently, the output is marked with labeled data from K-means clustering. Upon inspecting the visual representation, it becomes evident that the clustering is highly effective. Most of the "normal" blocks are tightly grouped together, while, in contrast, the "suspicious" blocks are dispersed, maintaining a noticeable separation from the "normal" ones.

7.4 RANDOM FOREST EVALUATION

Evaluation with Confusion Matrix

A "confusion matrix" serves as a valuable tool for assessing the performance of classification models, especially in supervised learning scenarios within machine learning and statistics. It's also referred to as an error matrix and offers a concise summary of a classification algorithm's actual and predicted outcomes. While typically applied in binary classification tasks, it can be adapted for multiclass classification as well.

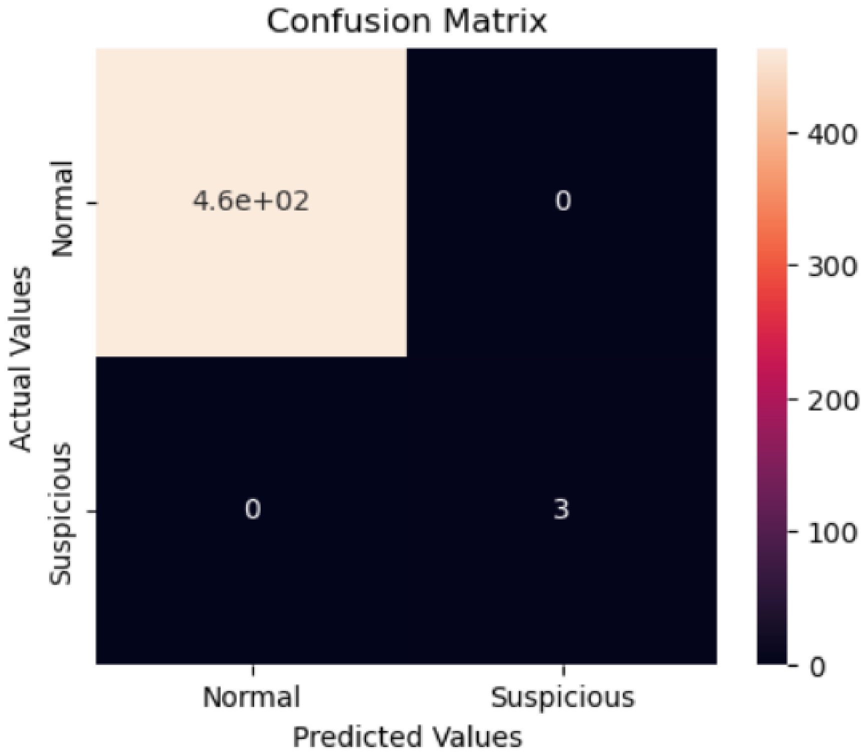


Figure 22: Confusion Matrix for Random Forest

In the context of this experiment involving 466 test data points for Bitcoin transaction blocks, the model demonstrates impressive accuracy. It accurately identifies 463 blocks as "Normal" and correctly classifies 3 blocks as "Suspicious." Importantly, there are no instances where the model makes errors by incorrectly predicting "Normal" (Type I error) or "Suspicious" (Type II error). As a result, the confusion matrix effectively evaluates the anomaly detection model for Bitcoin transaction blocks in this experiment, showcasing its excellent performance.

7.5 FURTHER RESEARCH SCOPE

This thesis focuses on Anomaly detection model of Bitcoin transaction blocks. Simultaneously, it has unveiled several avenues for further research and development in this dynamic and ever-evolving field. The following areas offer promising opportunities for future investigations:

Deep Research of Transaction Blocks: This experiment highlights on identifying the suspicious transaction blocks which leads to the opportunity of inventing more information into each block. Through this research we can finally identify the particular fraud transaction to restrict the path of Money Laundering in Cryptocurrency world.

Enhanced Anomaly Detection Techniques: While K-means clustering, Principal Component Analysis (PCA), and Random Forest have been applied successfully in this thesis, further research could explore the efficacy of other unsupervised and supervised machine learning algorithms. Comparative studies could help identify the most suitable techniques for various types of cryptocurrency transactions.

Real-time Anomaly Detection: The cryptocurrency market operates 24/7, and transactions occur in real time. Anomaly detection model of this research can delve into the development of real-time anomaly detection models that can adapt to the rapidly changing nature of the blockchain.

Incorporating External Data Sources: Anomaly detection model in this research can be modified in other features as well. Augmenting transaction data with external data sources, such as social media sentiment or economic indicators, could improve anomaly detection accuracy. Further research can explore the integration of these additional features.

Privacy-Preserving Techniques: With user anonymity being a hallmark of cryptocurrencies, future research can investigate privacy-preserving anomaly detection methods that respect users' privacy while still detecting suspicious activities.

Scalability and Big Data Challenges: The scalability of anomaly detection models is crucial as the volume of cryptocurrency transactions continues to grow. Exploring distributed and scalable approaches for handling big data in this context is a significant research area.

Regulatory Compliance: Given the increasing scrutiny of cryptocurrency transactions by regulators, further research could focus on developing anomaly detection models that aid in regulatory compliance and reporting.

Interoperability: As cryptocurrencies expand and new blockchain networks emerge, research into interoperability between different blockchain networks for anomaly detection could be essential.

User Education and Awareness: In addition to technical research, there is a need for educational initiatives and user awareness campaigns to help cryptocurrency users protect themselves from potential threats and understand the importance of secure transactions.

Benchmarking and Evaluation Metrics: Establishing standardized benchmark datasets and evaluation metrics for anomaly detection in cryptocurrency transactions can facilitate fair comparisons between different research approaches.

Economic and Market Impacts: Investigating the economic and market impacts of anomaly detection on cryptocurrencies, including its influence on investor behavior and market dynamics, could be a valuable research direction.

In conclusion, the field of anomaly detection in cryptocurrency transactions offers a plethora of exciting research opportunities. The dynamic and high-stakes nature of cryptocurrencies necessitates continuous innovation and adaptation in the realm of security and anomaly detection. Future research in these areas can contribute to a safer and more secure cryptocurrency ecosystem while advancing our understanding of digital asset transactions.

8 CHAPTER 8: CONCLUSION

In conclusion, this thesis has explored the intricate realm of cryptocurrency transactions, specifically focusing on Bitcoin, and employed machine learning techniques to detect anomalies within this domain. Anomaly detection in cryptocurrency transactions is of paramount importance due to the evolving landscape of digital assets and the emerging threats in the form of cybercrimes.

The primary methodology encompassed the application of unsupervised K-means clustering to label normal and anomaly blocks within Bitcoin transaction data. This approach revealed significant insights into the inherent clustering nature of cryptocurrency transactions, providing a robust foundation for further analysis.

To further refine the model and enhance its predictive capabilities, supervised algorithms, namely Principal Component Analysis (PCA) and Random Forest, were employed. PCA effectively reduced the dimensionality of the dataset while preserving vital information, rendering the data more interpretable. Random Forest, a powerful ensemble learning technique, was leveraged to strengthen the model's performance.

The comprehensive evaluation of the model indicated promising results, paving the way for future forecasting endeavours in the domain of cryptocurrency transactions. However, it is crucial to acknowledge the ever-evolving nature of cryptocurrencies, necessitating ongoing vigilance and adaptation of the anomaly detection model to counter emerging threats.

This research contributes to the broader field of digital asset security and paves the way for the development of more robust and proactive measures to safeguard the integrity of cryptocurrency transactions. As cryptocurrencies continue to gain prominence, the importance of effective anomaly detection methods cannot be overstated.

In conclusion, the thesis has successfully demonstrated the efficacy of machine learning in anomaly detection for cryptocurrency transactions, opening new avenues for research and advancements in the field. The findings presented herein lay the foundation for more resilient and adaptive security measures in the evolving landscape of cryptocurrency transactions.

BIBLIOGRAPHY:

Journal Article

Baek, H., Oh, J., Kim, C. Y., & Lee, K. (2019). A Model for Detecting Cryptocurrency Transactions with Discernible Purpose. *IEEE*. https://doi.org/10.1109/icufn.2019.8806126

Baumann, A., Fabian, B., & Lischke, M. (2014). Exploring the Bitcoin Network. *ResearchGate*. https://doi.org/10.5220/0004937303690374

Bolton, R. J., & Hand, D. (2001). Unsupervised profiling methods for fraud detection. *ResearchGate*. https://www.researchgate.net/publication/2407747_Unsupervised_Profiling_Methods_for_Fraud_Detection

Breunig, M., Kriegel, H., Ng, R. T., & Sander, J. (2000). LOF. *ResearchGate*. https://doi.org/10.1145/342009.335388

Carlozo, B. L. (2017, July 1). What is blockchain? *Journal of Accountancy*. https://www.journalofaccountancy.com/issues/2017/jul/what-is-blockchain.html

Chen, L., Özsu, M. T., & Oria, V. (2005). Robust and fast similarity search for moving object trajectories. *ResearchGate*. https://doi.org/10.1145/1066157.1066213

Cuesta-Albertos, J. A., Gordaliza, A., & Matrán, C. (1997). Trimmed k-means: an attempt to robustify quantizers. *Annals of Statistics*, *25*(2). https://doi.org/10.1214/aos/1031833664

Farren, D. (2016, November 11). *Low latency anomaly detection and Bayesian network prediction of anomaly likelihood*. arXiv.org. https://arxiv.org/abs/1611.03898

Gramoli, V. (2020). From blockchain consensus back to Byzantine consensus. *Future Generation Computer Systems*, *107*, 760–769. https://doi.org/10.1016/j.future.2017.09.023

Gu, Z., Lin, D., & Wu, J. (2022). On-chain analysis-based detection of abnormal transaction amount on cryptocurrency exchanges. *Physica A: Statistical Mechanics and Its Applications*, *604*, 127799. https://doi.org/10.1016/j.physa.2022.127799

Heilman, E. (2015). *Eclipse attacks on {Bitcoin's} {Peer-to-Peer} network*. https://www.usenix.org/conference/usenixsecurity15/technical-sessions/presentation/heilman

Huang, B., Liu, Z., Chen, J., Liu, A., Liu, Q., & He, Q. (2017a). Behavior pattern clustering in blockchain networks. *Multimedia Tools and Applications*, *76*(19), 20099–20110. https://doi.org/10.1007/s11042-017-4396-4

Huang, B., Liu, Z., Chen, J., Liu, A., Liu, Q., & He, Q. (2017b). Behavior pattern clustering in blockchain networks. *Multimedia Tools and Applications*, *76*(19), 20099–20110. https://doi.org/10.1007/s11042-017-4396-4

Index of /class/cs224w-2013/projects2013. (n.d.). https://snap.stanford.edu/class/cs224w-2013/projects2013

Kim, J., Nakashima, M., Fan, W., Wuthier, S., Zhou, X., Kim, I., & Chang, S. (2022). A machine learning approach to anomaly detection based on traffic monitoring for secure blockchain networking. *IEEE Transactions on Network and Service Management*, *19*(3), 3619–3632. https://doi.org/10.1109/tnsm.2022.3173598

Kumar, N., Singh, A. P., Handa, A., & Shukla, S. K. (2020). Detecting Malicious Accounts on the Ethereum Blockchain with Supervised Learning. In *Lecture Notes in Computer Science* (pp. 94–109). https://doi.org/10.1007/978-3-030-49785-9_7

Leskovec, J. (2006, March 27). *Graph evolution: densification and shrinking diameters*. arXiv.org. https://arxiv.org/abs/physics/0603229

Li, J., Gu, C., Wei, F., & Chen, X. (2019). A survey on blockchain anomaly detection using data mining techniques. In *Communications in computer and information science* (pp. 491–504). https://doi.org/10.1007/978-981-15-2777-7_40

Li, S., Yen, D. C., Lu, W., & Wang, C. (2012). Identifying the signs of fraudulent accounts using data mining techniques. *Computers in Human Behavior*, *28*(3), 1002–1013. https://doi.org/10.1016/j.chb.2012.01.002

Lorenz, J., Silva, M. I., Aparício, D., Ascensão, J. T., & Bizarro, P. (2020). Machine learning methods to detect money laundering in the bitcoin blockchain in the presence of label scarcity. *ICAIF '20: Proceedings of the First ACM International Conference on AI in Finance*. https://doi.org/10.1145/3383455.3422549

Martin, K. E., Rahouti, M., Ayyash, M., & Alsmadi, I. (2021). Anomaly detection in blockchain using network representation and machine learning. *Security and Privacy*, *5*(2). https://doi.org/10.1002/spy2.192

Monamo, P. M., Marivate, V., & Twala, B. (2016a). Unsupervised learning for robust Bitcoin fraud detection. *IEEE*. https://doi.org/10.1109/issa.2016.7802939

Monamo, P. M., Marivate, V., & Twala, B. P. (2016b). A Multifaceted Approach to Bitcoin Fraud Detection: Global and Local Outliers. *IEEE*. https://doi.org/10.1109/icmla.2016.0039

Morse, M. D., & Patel, J. M. (2007). An efficient and accurate method for evaluating time series similarity. *ResearchGate*. https://doi.org/10.1145/1247480.1247544

Patel, V., Pan, L., & Rajasegarar, S. (2020). Graph Deep learning based anomaly detection in Ethereum blockchain network. In *Lecture Notes in Computer Science* (pp. 132–148). https://doi.org/10.1007/978-3-030-65745-1_8

Pham, T. H., & Lee, S. (2016). Anomaly detection in Bitcoin network using unsupervised learning methods. *arXiv (Cornell University)*. https://doi.org/10.48550/arxiv.1611.03941

Pocher, N., Zichichi, M., Merizzi, F., Shafiq, M. Z., & Ferretti, S. (2023). Detecting anomalous cryptocurrency transactions: An AML/CFT application of machine learning-based forensics. *Electronic Markets*, *33*(1). https://doi.org/10.1007/s12525-023-00654-3

Podgorelec, B., Turkanović, M., & Karakatič, S. (2019). A Machine Learning-Based method for automated blockchain transaction signing including personalized anomaly detection. *Sensors*, *20*(1), 147. https://doi.org/10.3390/s20010147

Reid, F. (2011, July 22). *An analysis of anonymity in the bitcoin system*. arXiv.org. https://arxiv.org/abs/1107.4524

Sarkar, S. K., Midi, H., & Rana, S. (2010). Detection of outliers and influential observations in Binary Logistic Regression: an Empirical study. *Journal of Applied Sciences*, *11*(1), 26–35. https://doi.org/10.3923/jas.2011.26.35

Signorini, M. (2018, July 10). *BAD: Blockchain Anomaly Detection*. arXiv.org. https://arxiv.org/abs/1807.03833

Su, Z., Khoshgoftaar, T. M., & Seliya, N. (2007). CLUSTERING-BASED NETWORK INTRUSION DETECTION. *International Journal of Reliability, Quality and Safety Engineering*, *14*(02), 169–187. https://doi.org/10.1142/s0218539307002568

Sunyaev, A. (2020). Distributed ledger technology. In *Springer eBooks* (pp. 265–299). https://doi.org/10.1007/978-3-030-34957-8_9

Vlachos, M., Kollios, G., & Gunopulos, D. (2003). Discovering similar multidimensional trajectories. *ResearchGate*. https://doi.org/10.1109/icde.2002.994784

Internet Articles / Webpages

Avcontentteam. (2023, September 13). *PCA | What is Principal component Analysis & How it works? (Updated 2023)*. Analytics Vidhya. https://www.analyticsvidhya.com/blog/2016/03/pca-practical-guide-principal-component-analysis-python/

Dhalla, A. (2022, January 7). The Math of Principal Component Analysis (PCA) - Analytics Vidhya - medium. *Medium*. https://medium.com/analytics-vidhya/the-math-of-principal-component-analysis-pca-bf7da48247fc

Dharmaraj. (2022, January 29). The math behind K-Means Clustering - Dharmaraj - Medium. *Medium*. https://medium.com/@draj0718/the-math-behind-k-means-clustering-4aa85532085e#:~:text=K%2DMeans%20Clustering%20is%20an,ten%20clusters%2C%20and%20so%20on.

Donges, N. (2021, July 22). *Random Forest: A complete guide for machine learning*. Built In. https://builtin.com/data-science/random-forest-algorithm

Dubey, A. (2022, September 18). The mathematics behind principal component analysis. *Medium*. https://towardsdatascience.com/the-mathematics-behind-principal-component-analysis-fff2d7f4b643

Gajawada, S. K. (2023, May 31). K-Means Clustering with Math - Towards Data Science. *Medium*. https://towardsdatascience.com/k-means-clustering-for-beginners-2dc7b2994a4

Jaadi, Z. (2021, April 1). *A Step-by-Step Explanation of Principal Component Analysis (PCA)*. Built In. https://builtin.com/data-science/step-step-explanation-principal-component-analysis

K-Means Clustering | Brilliant Math & Science Wiki. (n.d.). https://brilliant.org/wiki/k-means-clustering/

K-Means Clustering — The Science of Machine Learning. (n.d.). The Science of Machine Learning. https://www.ml-science.com/k-means-clustering

Kumar, A. (2021, June 21). *KMEANs silhouette score explained with Python example*. dzone.com. https://dzone.com/articles/kmeans-silhouette-score-explained-with-python-exam

Mathematics behind K-Mean Clustering algorithm – Muthukrishnan. (2018, July 7). https://muthu.co/mathematics-behind-k-mean-clustering-algorithm/

Rastogi, R. (2021, December 14). Random Forest Classification and it’s Mathematical Implementation. *Medium*. https://medium.com/analytics-vidhya/random-forest-classification-and-its-mathematical-implementation-1895a7bb743e

S, R. A. (2023, October 18). *What is Blockchain Technology? How Does Blockchain Work? [Updated]*. Simplilearn.com. https://www.simplilearn.com/tutorials/blockchain-tutorial/blockchain-technology

Samariya, M. (2021, December 15). The mathematics behind Principal Component Analysis (PCA). *Medium*. https://medium.com/analytics-vidhya/mathematics-behind-principal-component-analysis-pca-1cdff0a808a9

Sumanth, K. (2023, March 31). *PCA And It’s Underlying Mathematical Principles*. Analytics Vidhya. https://www.analyticsvidhya.com/blog/2021/09/pca-and-its-underlying-mathematical-principles/

The surprising history of distributed ledger technology. (n.d.). https://www.knnx.com/blog/the-surprising-history-of-distributed-ledger-technology-133168

Code Related References

33. Random Forests in Python | Machine Learning | python-course.eu. (n.d.). https://python-course.eu/machine-learning/random-forests-in-python.php

Babitz, K. (2023, March 10). *Introduction to k-Means Clustering with scikit-learn in Python*. https://www.datacamp.com/tutorial/k-means-clustering-python

Brownlee, J. (2021, April 26). *How to develop a random forest ensemble in Python*. MachineLearningMastery.com. https://machinelearningmastery.com/random-forest-ensemble-in-python/

Educative. (n.d.). *Educative Answers - trusted answers to developer questions*. https://www.educative.io/answers/random-forest-in-python

Example of K-Means clustering in Python. (2023, October 31). https://datatofish.com/k-means-clustering-python/

Galarnyk, M. (2022, November 29). *PCA using Python: A tutorial*. Built In. https://builtin.com/machine-learning/pca-in-python

GeeksforGeeks. (2023, April 22). *Principal Component Analysis with Python*. https://www.geeksforgeeks.org/principal-component-analysis-with-python/

Keita, Z. (2023, February 24). How to perform KMEANs Clustering using Python - towards Data Science. *Medium*. https://towardsdatascience.com/how-to-perform-kmeans-clustering-using-python-7cc296cec092

Khan, A., & Khan, A. (2023, July 8). *Building random forest algorithm models in Python and Sklearn*. Intellipaat Blog. https://intellipaat.com/blog/what-is-random-forest-algorithm-in-python/

Online, S. (2023, August 23). *Random Forest Algorithm: Python Code - shiksha online*. Shiksha.com. https://www.shiksha.com/online-courses/articles/random-forest-algorithm-python-code/

Prashant. (2020a, March 13). *Random Forest Classifier tutorial*. Kaggle. https://www.kaggle.com/code/prashant111/random-forest-classifier-tutorial

Prashant. (2020b, December 8). *K-Means Clustering with Python*. Kaggle. https://www.kaggle.com/code/prashant111/k-means-clustering-with-python

Python, R. (2023, August 4). *K-Means Clustering in Python: A Practical Guide*. https://realpython.com/k-means-clustering-python/

Random Forest in Python with scikit-learn. (n.d.). datacareer.de. https://www.datacareer.de/blog/random-forest-in-python-with-scikit-learn/

Rogel-Salazar, J. (2023, September 1). Getting started with k-means clustering in Python. *Domino*. https://domino.ai/blog/getting-started-with-k-means-clustering-in-python

Sampaio, C. (2023, April 18). *Definitive Guide to the Random Forest Algorithm with Python and Scikit-Learn*. Stack Abuse. https://stackabuse.com/random-forest-algorithm-with-python-and-scikit-learn/

Sharma, A. (2020, January 1). *Principal Component Analysis (PCA) in Python tutorial*. https://www.datacamp.com/tutorial/principal-component-analysis-in-python

Sharma, P. (2023, August 9). *K means clustering | Step-by-Step tutorials for clustering in data analysis*. Analytics Vidhya. https://www.analyticsvidhya.com/blog/2021/04/k-means-clustering-simplified-in-python/

Sklearn.cluster.KMeans. (n.d.). Scikit-learn. https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html

Sklearn.decomposition.PCA. (n.d.). Scikit-learn. https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html

Towards AI Editorial Team. (2021, December 16). Principal Component Analysis (PCA) with Python Examples — Tutorial. *Medium*. https://pub.towardsai.net/principal-component-analysis-pca-with-python-examples-tutorial-67a917bae9aa

Ujhelyi, T. (2023, March 3). *Random Forest in Python and coding it with Scikit-learn (tutorial)*. Data36. https://data36.com/random-forest-in-python/

VanderPlas, J. (n.d.-a). *In depth: Principal Component Analysis | Python Data Science Handbook*. https://jakevdp.github.io/PythonDataScienceHandbook/05.09-principal-component-analysis.html

VanderPlas, J. (n.d.-b). *In-Depth: Decision Trees and Random Forests | Python Data Science Handbook*. https://jakevdp.github.io/PythonDataScienceHandbook/05.08-random-forests.html

Zyprian, F. (2023, October 5). *5 Wege zur Implementierung eines Random Forest in Python - Developer Guide*. Konfuzio. https://konfuzio.com/en/random-forest/

APPENDIX: PROGRMMING AND CODING DETAILS

Importing Libraries

```
# Importing necessary library and packages
# General Use
import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
import numpy as np
from sklearn.preprocessing import MinMaxScaler
from sklearn.preprocessing import StandardScaler

# Clustering
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

# 3D Visualization
import plotly as py
import plotly.graph_objs as go

# Principal Component Analysis
from sklearn.decomposition import PCA

# Random Forest
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import confusion_matrix
# Set random seed
np.random.seed(0)
```

Data Preparation

```
#Importing the data of block information
data1 = pd.read_excel("blockhash.xlsx")
data1.head()
```

```
#Importing the data of transaction
data2 = pd.read_excel("tx.xlsx")
data2.head()
```

```
#Importing the data of transaction inputs
data3 = pd.read_excel("txin.xlsx")
data3.head()
```

```
#Importing the data of transaction outputs
data4 = pd.read_excel("txout.xlsx")
data4
```

```
#Merging the dataset data2 and data3
df1 = pd.merge(data2, data3, on="txID")
df1.head()
```

```
#Merging the dataset df1 and data4
df2 = pd.merge(df1, data4, on="txID")
df2.head()
```

```
#Creating pivot table to get the information of blocks
pivot = pd.pivot_table(df2, values=['n_inputs', 'n_outputs', 'value_x', 'value_y'], index=['blockID'],
                       aggfunc=np.sum, fill_value=0)
pivot.reset_index()
pivot.head()
```

```
#Merging the dataset pivot and data and preparing the final dataset
final = pd.merge(pivot, data1, on="blockID")
final = final.rename(columns={'value_x': 'send_value', 'value_y': 'receive_value'})
final.drop(columns=['bhash'], inplace=True)
final['blockID'] = final['blockID'].astype(str)
final.head()
```

```
#the datatypes of final dataset
final.info()
```

Exloratory Data Analysis

Understanding the data

```
#the head of final dataset
final.head()
```

```
#the tail of final dataset
final.tail()
```

```
#the shape of final dataset
final.shape
```

```
#the summary statistics of final dataset
final.describe()
```

```
#the columns of final dataset
final.columns
```

```
#the number of unique values of different variable
final.nunique()
```

Cleaning the data

```
#total null values of different variables
final.isnull().sum()
```

```
#dropping the dependent variable blockID
final = final.drop(['blockID'], axis=1)
final.head()
```

Relationship analysis

```
#setting the correlation coefficient
correlation = final.corr()
```

```
#Using heatmap to illustrate the correlations between variables
sns.heatmap(correlation, xticklabels=correlation.columns, yticklabels=correlation.columns, annot=True)
```

```
#Using pairplot to illustrate the correlations between variables
sns.pairplot(final)
```

```
#Using distribution plot to illustrate the distribution of n_inputs
sns.distplot(final["n_inputs"])
```

```
#Using distribution plot to illustrate the distribution of n_outputs
sns.distplot(final["n_outputs"])
```

```
#Using distribution plot to illustrate the distribution of send_value
sns.distplot(final["send_value"])
```

```
#Using distribution plot to illustrate the distribution of recieve_value
sns.distplot(final["receive_value"])
```

```
#Using distribution plot to illustrate the distribution of block creation time
sns.distplot(final["btime"])
```

```
#Using distribution plot to illustrate the distribution of transactions
sns.distplot(final["txs"])
```

```
#Using box plot to illustrate the distribution of n_inputs
sns.catplot(x='n_inputs', kind='box',data=final)
```

```
#Using boxplot plot to illustrate the distribution of n_outputs
sns.catplot(x='n_outputs', kind='box',data=final)
```

```
#Using box plot to illustrate the distribution of send_value
sns.catplot(x='send_value', kind='box',data=final)
```

```
#Using box plot to illustrate the distribution of receive_value
sns.catplot(x='receive_value', kind='box', data=final)
```

```
#Using box plot to illustrate the distribution of block creation time
sns.catplot(x='btime', kind='box', data=final)
```

```
#Using box plot to illustrate the distribution of transactions
sns.catplot(x='txs', kind='box', data=final)
```

K-means Clustering

```
#Preparaing the dataset for K-means algorithm
dataD = final[["n_inputs", "n_outputs", "send_value", "receive_value", "btime", "txs"]]
dataD
```

Finding the Optimal Number of Clusters with the Elbow Method

```
#Setting the algorithm for the Elbow Method
sum_of_sqr_dist = {}

for k in range(1, 10):
    km = KMeans(n_clusters=k, init='k-means++', max_iter=1000)
    km = km.fit(dataD)
    sum_of_sqr_dist[k] = km.inertia_
```

```
#Visualizing the Elbow Method for Optimal number of clustering
sns.pointplot(x=list(sum_of_sqr_dist.keys()), y=list(sum_of_sqr_dist.values()))
plt.xlabel("Number of Clusters (K)")
plt.ylabel("Sum of Square Distances")
plt.title("Elbow Method for Optimal K")
plt.show()
```

```
#Setting the KMeans algorithm for the number of clusters 2
Model2 = KMeans(n_clusters=2, init='k-means++', max_iter=1000)
Model2.fit(dataD)
```

```
#Fitting the model and visualizing the predicted clusters
dataD['Cluster'] = Model2.fit_predict(dataD)
dataD.head()
```

```
#Separating Normal blocks (Cluster==0) and Suspicious blocks (Cluster==1)
clx0 = dataD[dataD.Cluster==0]
clx1 = dataD[dataD.Cluster==1]
```

```
#the head of Suspicious blocks
clx1.head()
```

```
#Counting the number of blocks in two clusters
kmeans_model = KMeans(n_clusters = 2, random_state = 1).fit(dataD)
dataD['kmean'] = kmeans_model.labels_
dataD['kmean'].value_counts()
```

Evaluation with Silhouette Score

```
#Setting the labels variable
labels = Model2.labels_
```

```
#the silhouette score for this dataset
silhouette_score(dataD, labels)
```

```
#Setting the algorithm for the silhouette score of different number of clusters
silhouette = {}

for k in range(2,8):
    km = KMeans(n_clusters=k, init='k-means++', max_iter=1000)
    km.fit(dataD)
    silhouette[k] = silhouette_score(dataD, km.labels_)
```

```
#Visualizing the silhouette score of different number of clusters
sns.pointplot(x=list(silhouette.keys()), y=list(silhouette.values()))
plt.xlabel("Number of Clusters (K)")
plt.ylabel("Silhouette Scores")
plt.title("Silhouette Scores for Each K")
plt.show()
```

Principal Component Analysis

```
#the head of dataset for Principal Component Analysis
dataD.head()
```

```
#Preparing the dataset for Principal Component Analysis
PCAD = pd.merge(final, dataD, on=["n_inputs", "n_outputs", "receive_value", "send_value", "btime", "txs"])
FPCAD = PCAD['Cluster']
PCAD.drop(columns=['Cluster', 'kmean'], inplace=True)
PCAD
```

```
#the columns of the dataset
PCAD.keys()
```

```
#Scaling the data StandardScaler
scaler = StandardScaler()
scaler.fit(PCAD)
scaled_data = scaler.transform(PCAD)
```

```
#Setting the algorithm for Principal Component Analysis
pca = PCA(n_components = 2)
pca.fit(scaled_data)
```

```
#The dimensionality reduction with Principal Component Analysis
x_pca = pca.transform(scaled_data)
print(scaled_data.shape)
print(x_pca.shape)
```

```
#Visualizing the Principal Components of Principal Component Analysis
plt.figure(figsize=(8,6))
plt.scatter(x_pca[:,0], x_pca[:,1], c=FPCAD, cmap='plasma')
plt.xlabel('First principal component')
plt.ylabel('Second principal component')
```

```
#Interpreting the components
pca.components_
```

```
#Visualizing the contributions of the components
df_comp = pd.DataFrame(pca.components_, columns = PCAD.keys())
plt.figure(figsize=(12,6))
sns.heatmap(df_comp, cmap='plasma')
```

Random Forest

```
#Preparing the dataset for Random Forest
dataD.drop(columns=['kmean'], inplace=True)
dataD
```

```
#Creating Test and Train Data
dataD['is_train'] = np.random.uniform(0, 1, len(dataD)) <= .75

#the head of the dataset
dataD.head()
```

```
#Creating dataframes with test rows and training rows
train, test = dataD[dataD['is_train']==True], dataD[dataD['is_train']==False]

#Showing the number of observations for the test and training dataframes
print('Number of observations in the training data:', len(train))
print('Number of observations in the test data:',len(test))
```

```
#Creating a list of the feature column's names
features = dataD.columns[:6]

#Viewing features
print(features)
```

```
#Viewing target
y = train['Cluster']
y.head()
```

```
#Creating a random forest Classifier.
clf = RandomForestClassifier(n_jobs=2, random_state=0)

#Training the Classifier
clf.fit(train[features], y)
```

```
#Applying the trained classifier to the test
clf.predict(test[features])
```

```
#Viewing the predicted probabilities of the first 10 observations
clf.predict_proba(test[features])[0:10]
```

```
#Creating actual english names for the plants for each predicted plant class
preds = clf.predict(test[features])
preds = pd.Series(np.where(preds == 1, 'Suspicious', 'Normal'))

#Viewing the PREDICTED species
preds.head()
```

```
#Viewing the ACTUAL species for the first five observations
actual = test['Cluster']
actual = pd.Series(np.where(actual == 1, 'Suspicious', 'Normal'))
actual.head()
```

Evaluation with Confusion Matrix

```
#Settting up the confusion matrix
cm = confusion_matrix(actual, preds)
cm
```

```
#Creating a dataframe for a array-formatted Confusion matrix,so it will be easy for plotting
cm_df = pd.DataFrame(cm,
                     index = ['Normal','Suspicious'],
                     columns = ['Normal','Suspicious'])
```

```
#Plotting the confusion matrix
plt.figure(figsize=(5,4))
sns.heatmap(cm_df, annot=True)
plt.title('Confusion Matrix')
plt.ylabel('Actual Values')
plt.xlabel('Predicted Values')
plt.show()
```

DECLARATION OF AUTHENTICITY

I hereby declare that, the Master's thesis conducted independently and used only listed materials and sources are used. All materials used, from published as well as unpublished sources, whether directly quoted or paraphrased, are duly reported.

Furthermore, I declare that the Master's thesis, or any abridgment of it, was not used for any other degree seeking purpose.

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